Machine Learning via Azure

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot | labuser24@helloaicloud... HELLOAILAB (HELLOAICLOUD.O...

Home > rg-labuser24 > Marketplace >

Azure Machine Learning

Microsoft

Azure Machine Learning

Add to Favorites

Microsoft | Azure Service

★ 4.3 (155 ratings)

Azure benefit eligible

Subscription: Hands on lab

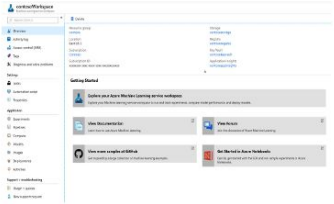
Plan: Azure Machine Learning

Create

Overview | Plans | Usage Information + Support | Ratings + Reviews

Azure Machine Learning empowers developers and data scientists with a wide range of productive experiences for building, training, and deploying machine learning models.
Create an Azure Machine Learning workspace to train, manage, and deploy machine-learning experiments and web services.

Media



More products from Microsoft See All

Give feedback

https://portal.azure.com/#

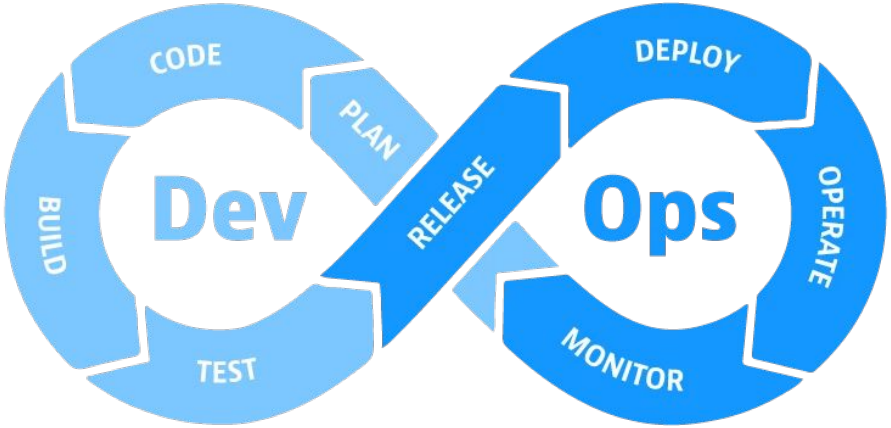
Normally we need to register to use GPU's but for today we just use CPU

After creating/deploying the resource, go to the **machine learning studio**.

정확도 92% 짜리는 cpu를 40%만 차지하고 95%는 cpu를 80% 넘게 차지한다? → 92%를 쓰는게 맞음.

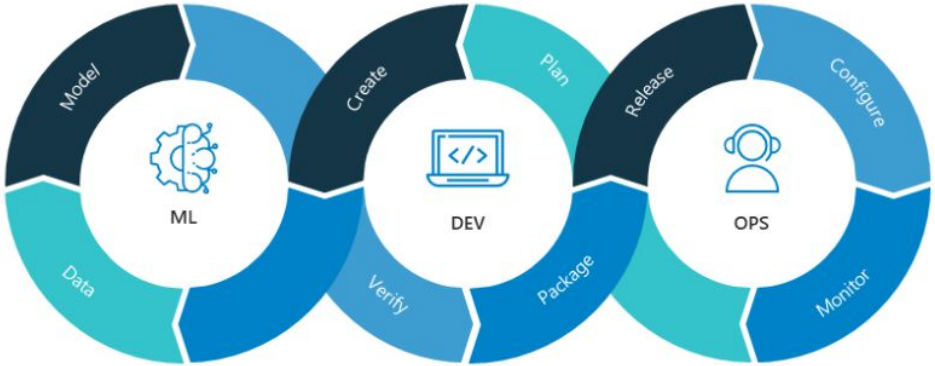
적당한 가격과 적당한 모델을 취사선택해서 빨리 성과를 내는게 중요. 그래서 **data scientist보다 engineer가 더 중요해짐**.

모델 선택, 클라우드, 배포, 관리 등, 전체 프로세스를 관리하는 엔지니어링 영역이 더 중요하다 (DevOps!).



머신러닝 쪽 특화된 프로세스 = MLOps?
e.g. 중고차 시장: 한번 만들어 놓은 모델로 중고차 가격을 계속 예측할 수 없다?

2026년 기준 MLOps(Machine Learning Operations)는 인공지능 모델의 개발(Dev)과 운영(Ops)을 통합하여, 머신러닝 생태계 전반을 관리하고 자동화하는 핵심 실무 체계입니다.



모든 작업 영역

홈

모델 카탈로그

작성

Notebooks

자동화된 ML

디자이너

프롬프트 흐름

자산

데이터

작업(Job)

구성 요소

파이프라인

환경

모델

엔드포인트

관리

컴퓨팅

모니터링

데이터 레이블 지정

연결

컴퓨팅

이제 "Kubernetes 클러스터" 탭에서 해당 유형을 사용하여 이전에 만든 컴퓨팅 대상과 함께 이전 버전의 "유추 클러스터"("AKS 클러스터"라고도 함)와 "연결된 Kubernetes" 컴퓨팅 유형에 액세스할 수 있습니다. 자세히 알아보기 Kubernetes 클러스터 정보.

컴퓨팅 인스턴스 | 컴퓨팅 클러스터 | Kubernetes 클러스터 | 연결된 컴퓨팅 | 서버리스 인스턴스

VS Code, JupyterLab, Jupyter, RStudio, ML 패키지, 딥 러닝 프레임워크 및 GPU 드라이버와 같은 널리 사용되는 도구로 미리 구성된 다양한 CPU 또는 GPU 인스턴스 중에서 선택하세요. 컴퓨팅 인스턴스에 대해 자세히 알아보기

컴퓨팅 인스턴스:
basically a virtual machine



컴퓨팅 인스턴스를 만들어 Azure Machine Learning Notebook 및 R 스크립트 시작

VS Code, JupyterLab, Jupyter 및 RStudio, ML 패키지, 딥 러닝 프레임워크 및 GPU 드라이버와 같은 널리 사용되는 도구로 사전 구성된 CPU 또는 GPU 인스턴스 중에서 선택합니다. 자세한 정보

+ 새로 만들기

Azure Machine Learning 자습서 보기
사용 가능한 할당량 보기

컴퓨팅 인스턴스 만들기

1 필수 설정
2 일정 예약 선택적
3 보안 선택적
4 애플리케이션 선택적
5 태그 선택적
6 검토

컴퓨팅 인스턴스에 사용할 이름 및 가상 머신 크기를 선택합니다.

ⓘ 컴퓨팅 인스턴스는 공유할 수 없습니다. 할당된 단일 사용자만 사용할 수 있습니다. 기본적으로 작성자에게 할당되며 보안 단계에서 다른 사용자로 변경할 수 있습니다.

컴퓨팅 이름 * ⓘ

labuser241

가상 머신 유형 ⓘ

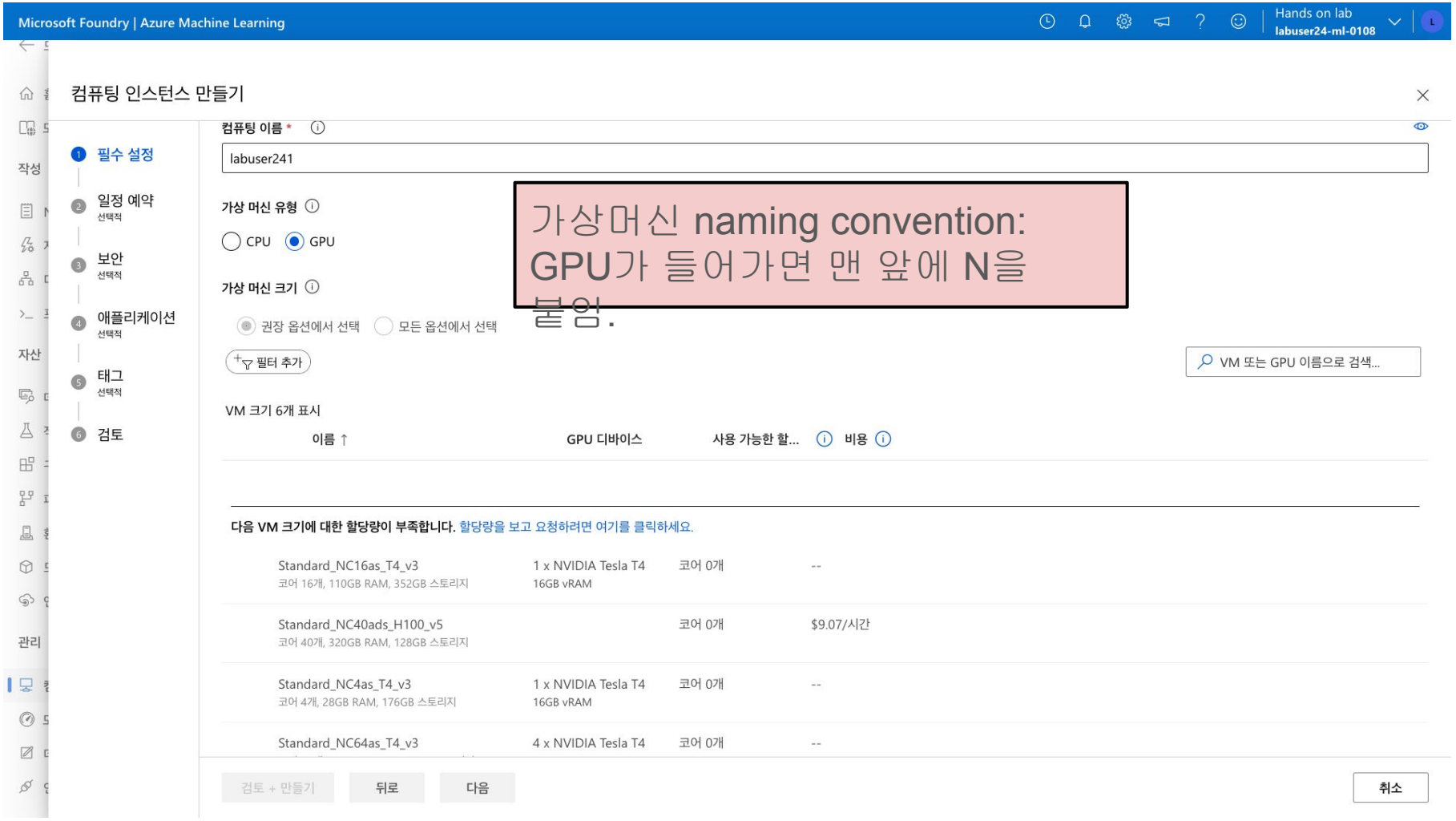
(●) CPU ( ) GPU

가상 머신 크기 ⓘ

(●) 권장 옵션에서 선택 ( ) 모든 옵션에서 선택

	이름 ↑	범주	워크로드 유형	사용 가능한 할... ⓘ	비용 ⓘ
( )	Standard_DS11_v2 코어 2개, 14GB RAM, 28GB 스토리지	메모리 최적화	Notebook(또는 다른 IDE)에서 개발 및 경량 테스트	코어 6개	$0.17/시간
( )	Standard_DS3_v2 코어 4개, 14GB RAM, 28GB 스토리지	범용	작은 데이터 세트에 대한 클래식 ML 모델 학습	코어 6개	$0.27/시간
(✓)	Standard_E4s_v3 코어 4개, 32GB RAM, 64GB 스토리지	메모리 최적화	중간 크기의 데이터 세트에 대한 데이터 조작 및 학습(1~10GB)	코어 6개	$0.28/시간
( )	Standard_F4s_v2 코어 4개, 8GB RAM, 32GB 스토리지	컴퓨팅 최적화	큰 데이터 세트에 대한 데이터 조작 및 학습(>10GB)	코어 6개	$0.19/시간

검토 + 만들기 | 뒤로 | 다음 | 취소



가상머신 naming convention:
GPU가 들어가면 맨 앞에 N을 붙임.

CPU vs. GPU

GPU는 코어가 많아 Loop 연산을 병렬 처리합니다. CPU의 직렬 연산을 병렬로 전환하는 것이죠. 문제는 GPU에 맞춰 코딩하기 어렵다는 것인데, 이를 돕는 것이 CUDA입니다. (요즘은 구글 TPU 같은 탈-CUDA가 대세입니다.)

도전 과제: 한 스레드가 다른 스레드 결과에 의존하면 (예: 피보나치 수 계산) 병렬화가 극도로 어려워지므로, 의존성을 피하도록 수학적 설계를 재구성해야 합니다.

Quick Comparison

Feature	CPU	GPU
Cores	Few (4–64), but very powerful.	Thousands (1,000+), but smaller and weaker.
Strength	Complex logic, branching, and sequential steps.	Repetitive calculations and massive data throughput.
Best For	Running the OS, opening apps, general computing.	Gaming graphics, AI/Deep Learning, Video Rendering.

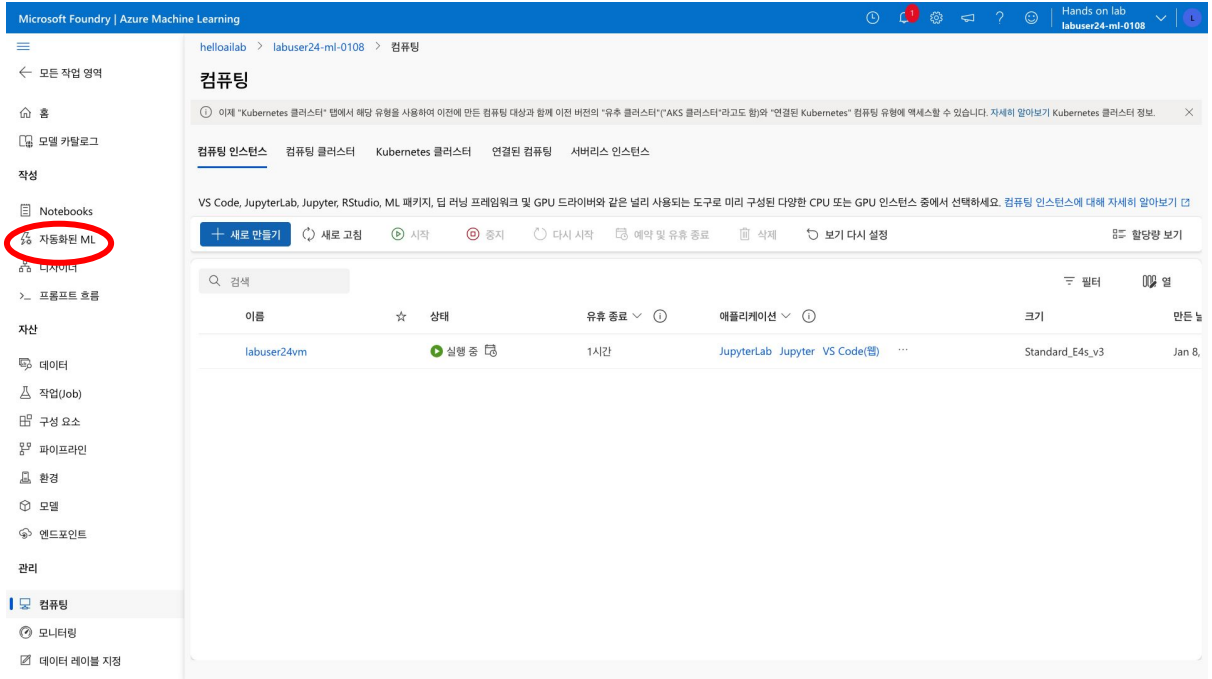
Sheets로 내보내기

Automated ML

ML의 전반적인 workflow를 살펴보면…

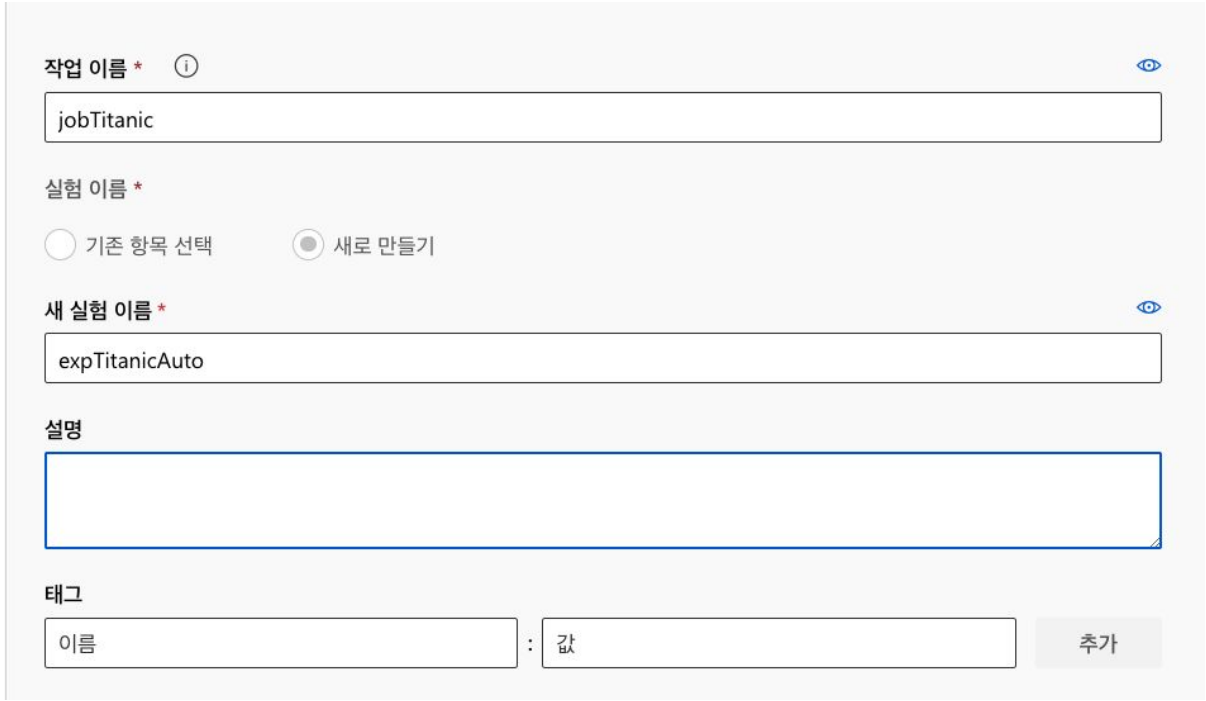
1. Data Preprocessing
2. Model training
3. Hyperparam tuning

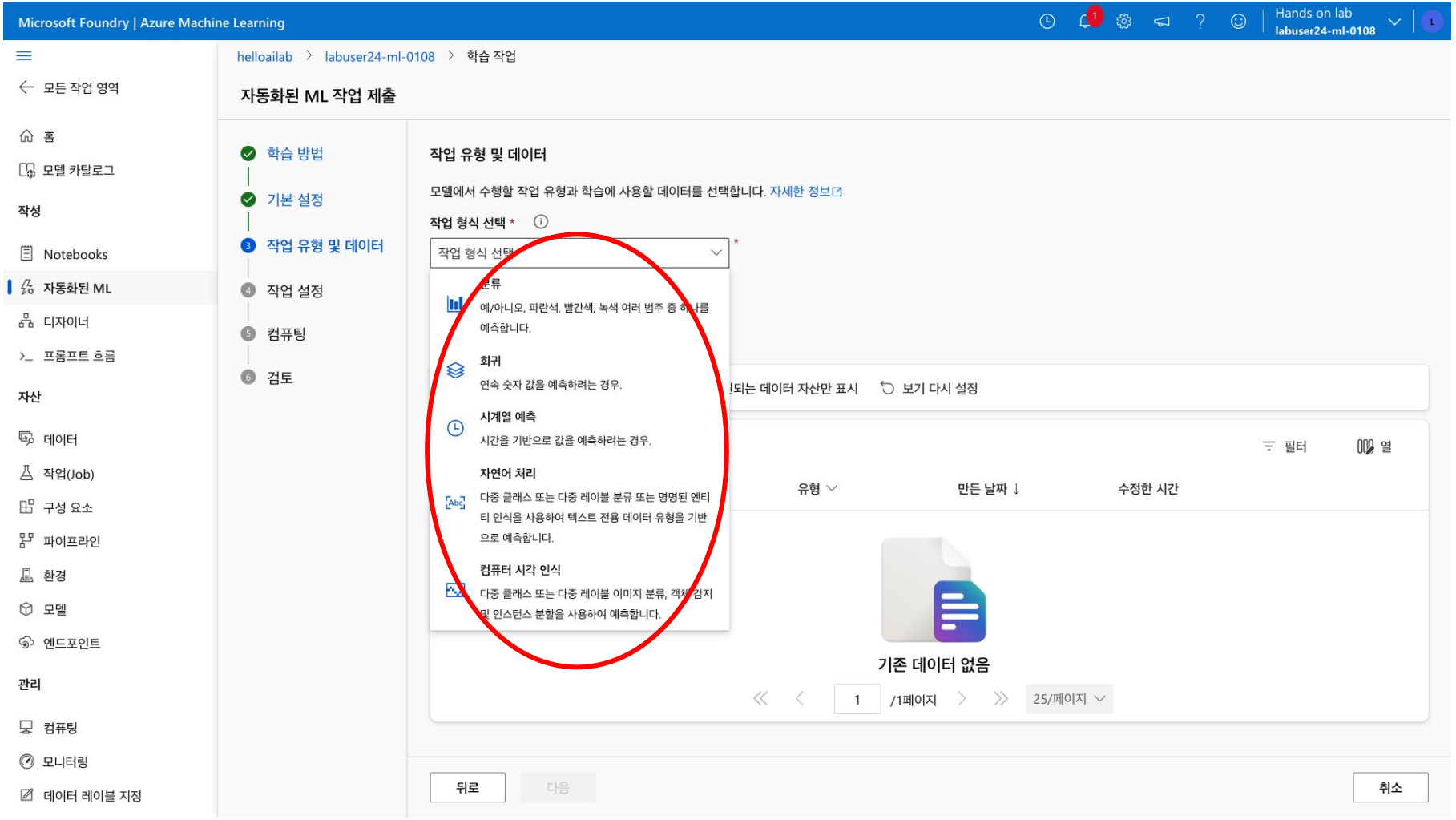
Automated Machine Learning이란,
데이터만 집어넣으면 1→2→3을 자동으로 해주는 것!



작업이름 앞에 job

실험이름 앞에 exp가 컨벤션.




Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
모든 작업 영역
홈
모델 카탈로그
작성
Notebooks
자동화된 ML
디자이너
프롬프트 흐름
자산
데이터
작업(Job)
구성 요소
파이프라인
환경
모델
엔드포인트
관리
컴퓨팅
모니터링
데이터 레이블 지정
helloailab
labuser24-ml-0108
학습 작업
자동화된 ML 작업 제출
학습 방법
기본 설정
작업 유형 및 데이터
작업 설정
컴퓨팅
검토
작업 유형 및 데이터
모델에서 수행할 작업 유형과 학습에 사용할 데이터를 선택합니다. 자세한 정보
작업 형식 선택 *
작업 형식 선택
분류
예/아니오, 파란색, 빨간색, 녹색 여러 범주 중 하나를 예측합니다.
회귀
연속 숫자 값을 예측하려는 경우.
시계열 예측
시간을 기반으로 값을 예측하려는 경우.
자연어 처리
다중 클래스 또는 다중 레이블 분류 또는 명명된 엔티티 인식을 사용하여 텍스트 전용 데이터 유형을 기반으로 예측합니다.
컴퓨터 시각 인식
다중 클래스 또는 다중 레이블 이미지 분류, 객체 감지 및 인스턴스 분할을 사용하여 예측합니다.
보기 다시 설정
필터
열
유형
만든 날짜
수정한 시간
기존 데이터 없음
1
/1페이지
25/페이지
뒤로
다음
취소

데이터 자산 만들기

1. 데이터 형식
2. 데이터 원본

데이터 자산의 원본 선택

자산을 만들 데이터 원본을 선택하세요. 데이터 원본은 컴퓨터의 로컬 스토리지 위치, 연결된 데이터 저장소, Azure Storage 또는 공개적으로 사용 가능한 웹 위치에서 가져올 수 있습니다.

Azure Storage에서

Azure Blob Storage, Azure 파일 공유, Azure Data Lake 등의 등록된 데이터 스토리지 서비스에서 데이터 자산을 만듭니다.

로컬 파일에서

로컬 드라이브에서 파일을 업로드하여 데이터 자산을 만듭니다.

SQL 데이터베이스에서

Azure SQL 데이터베이스 및 Azure PostGreSQL 데이터베이스에서 데이터 세트를 만듭니다.

웹 파일에서

공용 웹 URL에 있는 단일 파일에서 데이터 자산을 만듭니다.

Azure Open Datasets에서

미리 만들어진 데이터 세트에서 한 번의 클릭으로 데이터 세트를 만듭니다. 이러한 데이터 세트는 일반 대중이 생성하여 Azure Open Datasets로 게시됩니다.

DB와도 연동 가능하다!

뒤로 | 만들기 | 취소

titanic.csv를 올린후... 데이터 전처리 과정을 이렇게 interactive하게 할 수 있다.
심지어 결측치 제거도 알아서 해줌

Microsoft Foundry | Az

데이터 자산

- 데이터 형식
- 데이터 원본
- 대상 스토리지 유형
- 파일 또는 폴더 선택
- 설정
- 6 스키마
- 7 검토

작성 자산 관리

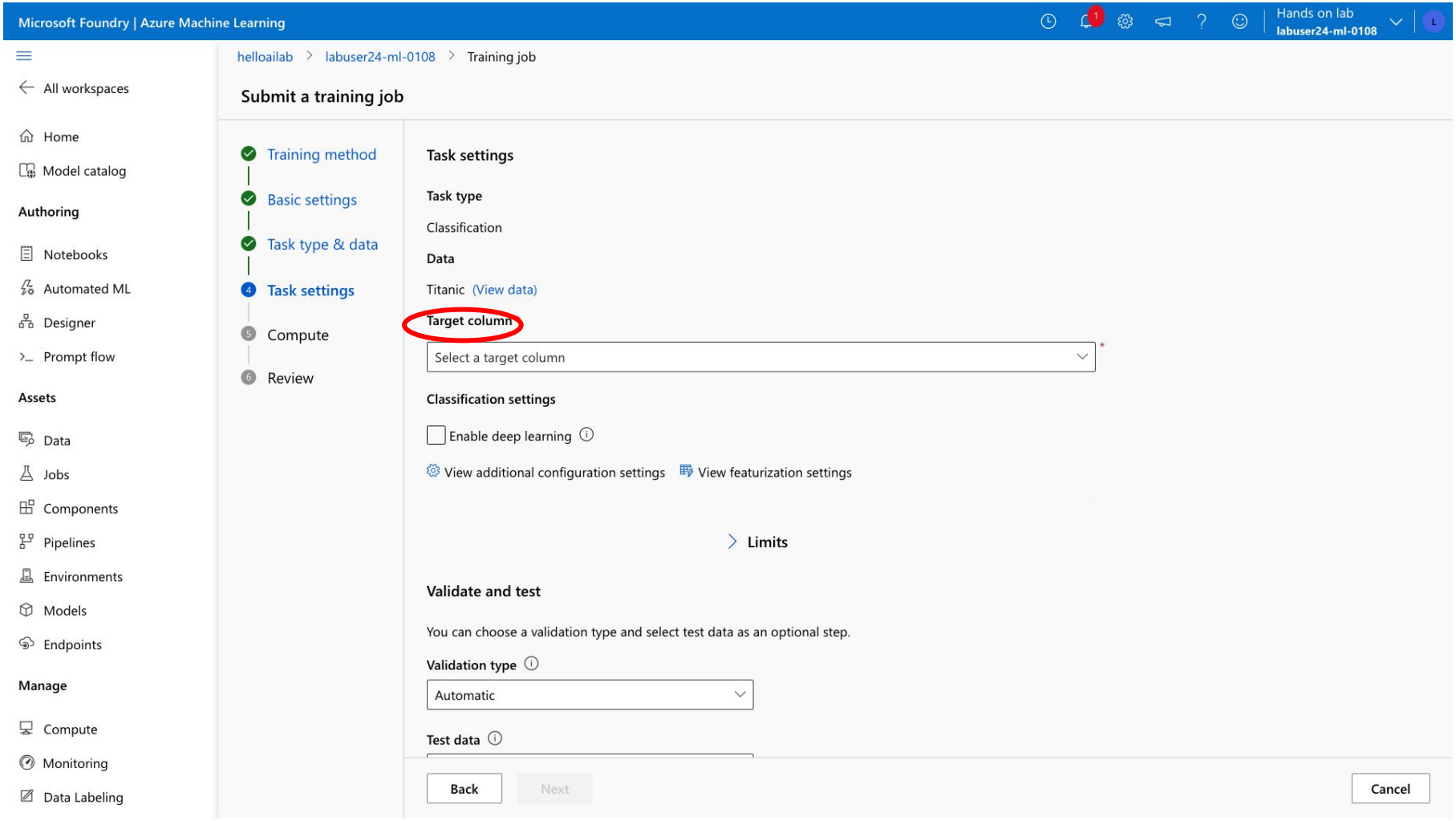
스키마

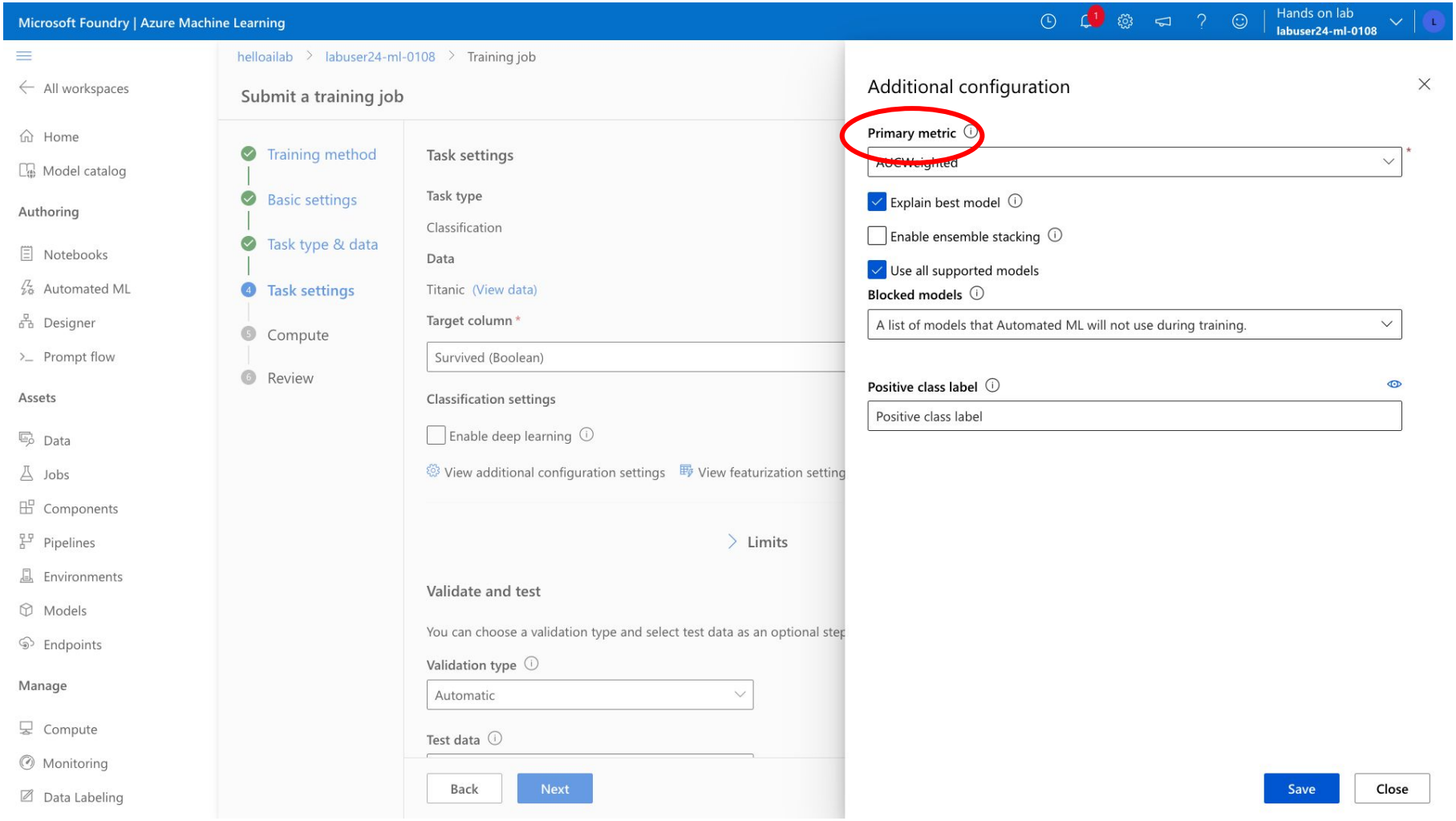
열 형식은 데이터의 초기 하위 집합을 기반으로 자동 검색되며 여기에서 업데이트할 수 있습니다. 지정한 열 형식에 맞지 않는 값은 변환에 실패하며 null로 채워지거나 오류 값으로 바뀔 수 있습니다. 변환 미리 보기 오류는 논블로킹이므로 계속 진행할 수 있습니다.

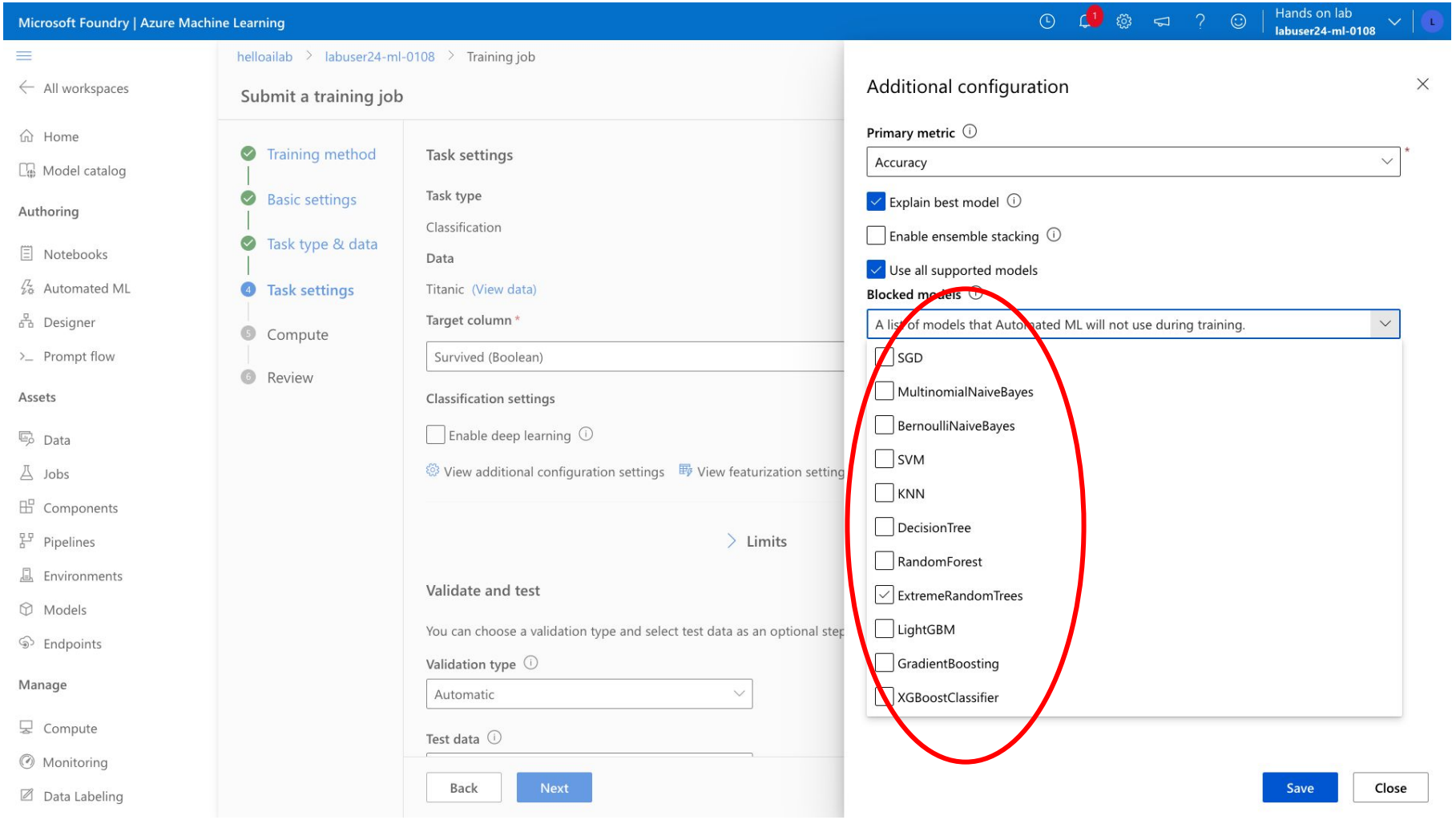
열 이름 검색

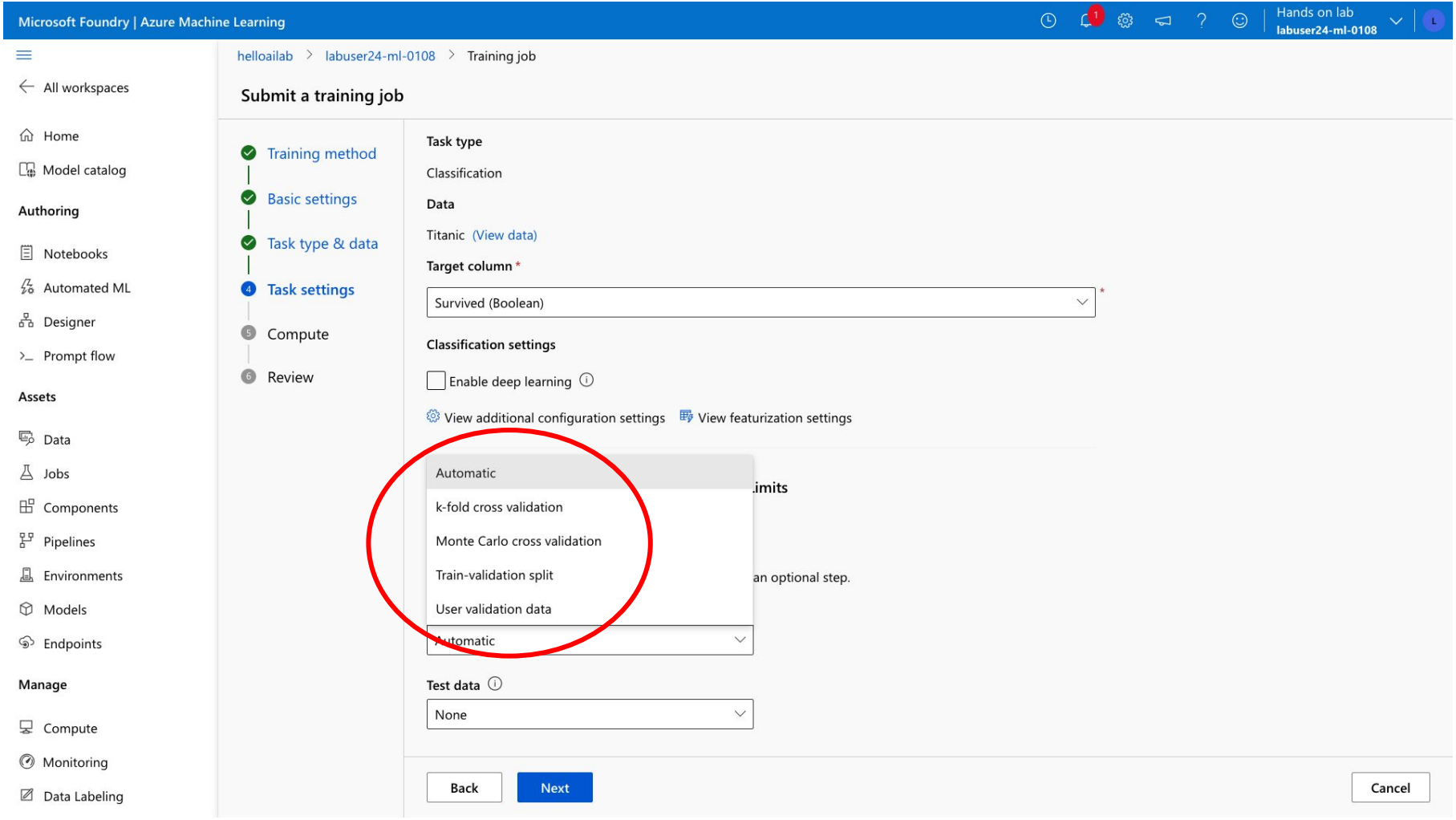
포함	열 이름	유형	예제 값	날짜 형식	속성
○	Path	문자열		선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	PassengerId	정수	1, 2, 3	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	Survived	정수	0, 1, 1	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	Pclass	정수	3, 1, 3	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	Name	문자열	Braund, Mr. Owen Harris, Cumings, Mr...	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	Sex	문자열	male, female, female	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	Age	소수 구분(점 '.')	22, 38, 26	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	SibSp	정수	1, 1, 0	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...
●	Parch	정수	0, 0, 0	선택한 유형에 적용할 수 없습니다.	선택한 유형에 적용할 ...

뒤로 다음 취소


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
helloailab
labuser24-ml-0108
Training job
Submit a training job
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Task settings
Task type
Classification
Data
Titanic (View data)
Target column
Select a target column
Classification settings
Enable deep learning
View additional configuration settings
View featurization settings
Limits
Validate and test
You can choose a validation type and select test data as an optional step.
Validation type
Automatic
Test data
Back
Next
Cancel


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
helloailab
labuser24-ml-0108
Training job
Submit a training job
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Task settings
Task type
Classification
Data
Titanic (View data)
Target column *
Survived (Boolean)
Classification settings
Enable deep learning
View additional configuration settings
View featurization setting
Limits
Validate and test
You can choose a validation type and select test data as an optional step
Validation type
Automatic
Test data
Back
Next
Additional configuration
Primary metric
AUCWeighted
Explain best model
Enable ensemble stacking
Use all supported models
Blocked models
A list of models that Automated ML will not use during training.
Positive class label
Positive class label
Save
Close


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
helloailab
labuser24-ml-0108
Training job
Submit a training job
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Task settings
Task type
Classification
Data
Titanic (View data)
Target column *
Survived (Boolean)
Classification settings
Enable deep learning
View additional configuration settings
View featurization setting
Limits
Validate and test
You can choose a validation type and select test data as an optional step
Validation type
Automatic
Test data
Back
Next
Additional configuration
Primary metric
Accuracy
Explain best model
Enable ensemble stacking
Use all supported models
Blocked models
A list of models that Automated ML will not use during training.
SGD
MultinomialNaiveBayes
BernoulliNaiveBayes
SVM
KNN
DecisionTree
RandomForest
ExtremeRandomTrees
LightGBM
GradientBoosting
XGBoostClassifier
Save
Close


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
helloailab
labuser24-ml-0108
Training job
Submit a training job
Training method
Basic settings
Task type & data
Task settings
Compute
Review
Task type
Classification
Data
Titanic (View data)
Target column *
Survived (Boolean)
Classification settings
Enable deep learning
View additional configuration settings
View featurization settings
Automatic
k-fold cross validation
Monte Carlo cross validation
Train-validation split
User validation data
Automatic
Test data
None
Back
Next
Cancel

All workspaces

Home
Model catalog

Authoring

Notebooks
Automated ML
Designer
Prompt flow

Assets

Data
Jobs
Components
Pipelines
Environments
Models
Endpoints

Manage

Compute
Monitoring
Data Labeling

Submit a training job

Training method
Basic settings
Task type & data
4 Task settings
5 Compute
6 Review

Task type

Classification

Data

Titanic (View data)

Target column *

Survived (Boolean) *

Classification settings

Enable deep learning

View additional configuration settings View featurization settings

> Limits

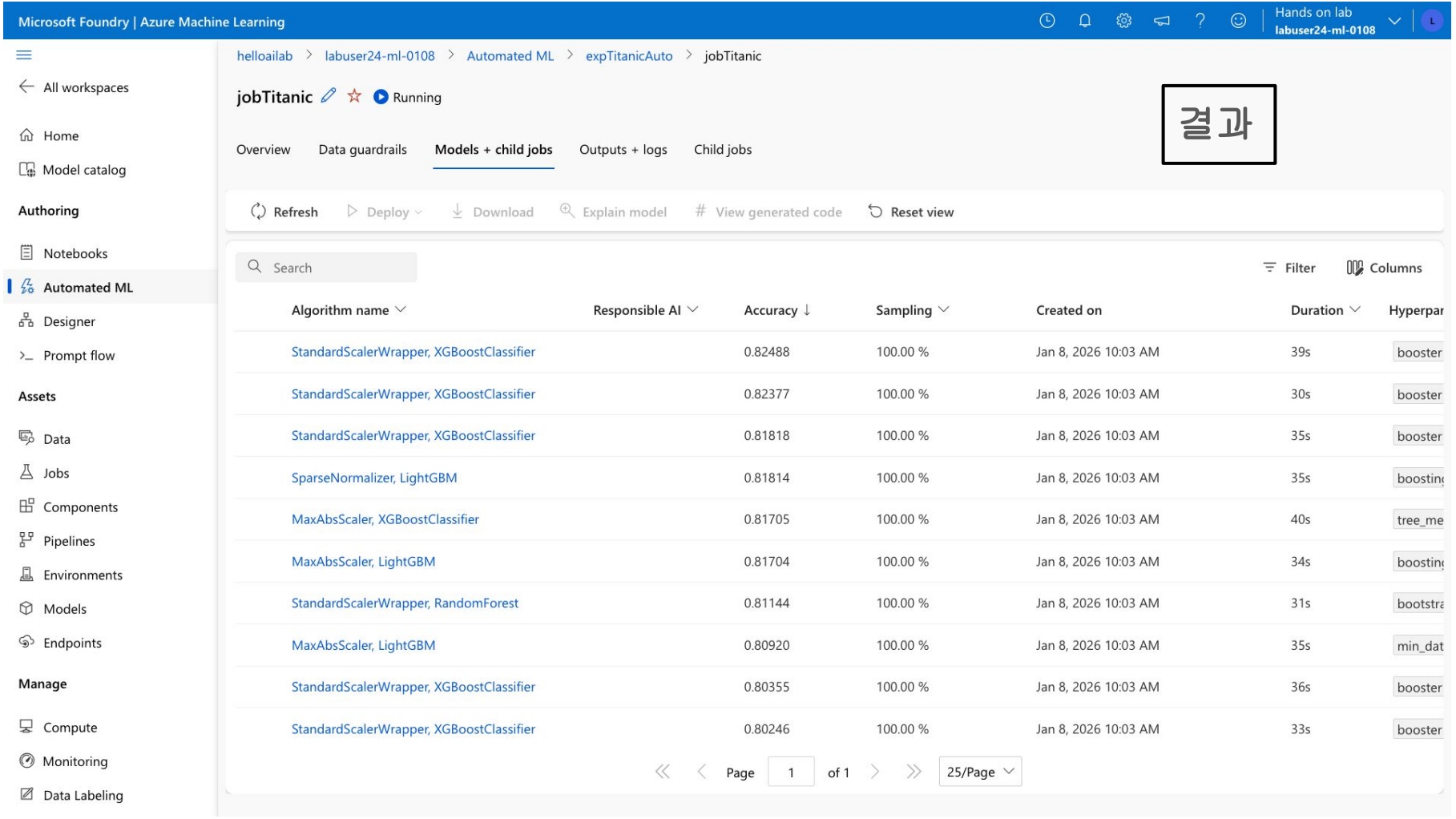
Validate and test

You can choose a validation type and select test data as an optional step.

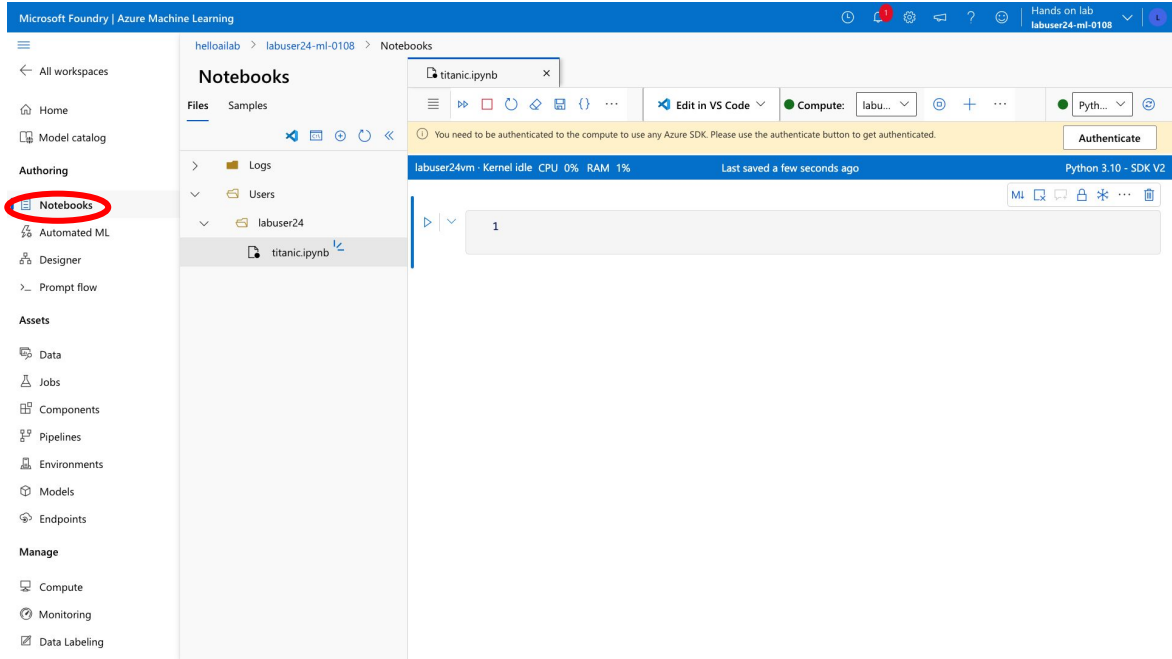
None
Train-test split
User test data

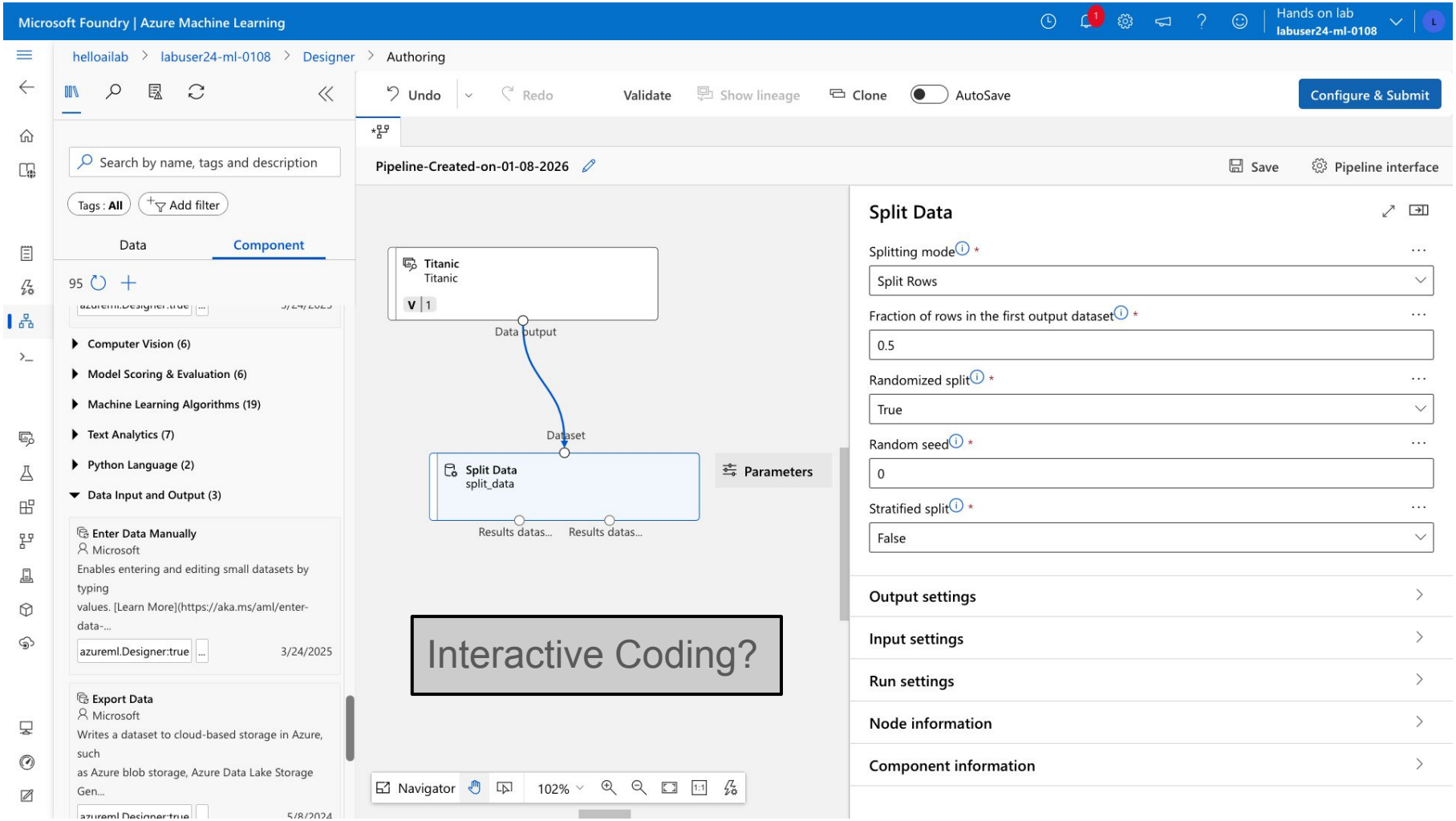
None

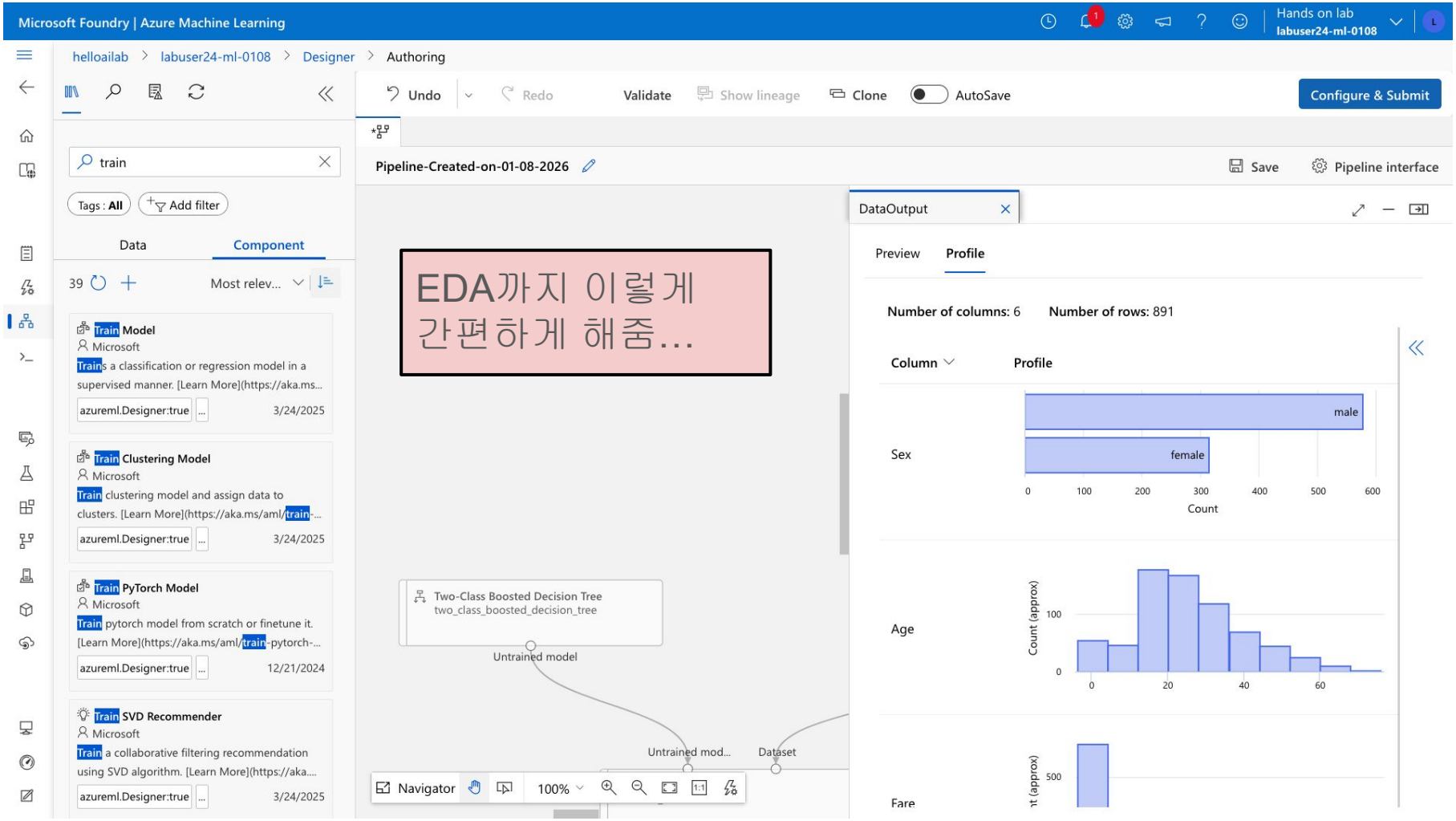
Back Next Cancel


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
helloailab > labuser24-ml-0108 > Automated ML > expTitanicAuto > jobTitanic
jobTitanic
Running
결과
Overview
Data guardrails
Models + child jobs
Outputs + logs
Child jobs
Refresh
Deploy
Download
Explain model
View generated code
Reset view
Search
Filter
Columns
Algorithm name
Responsible AI
Accuracy
Sampling
Created on
Duration
Hyperpar
StandardScalerWrapper, XGBoostClassifier 0.82488 100.00 % Jan 8, 2026 10:03 AM 39s booster
StandardScalerWrapper, XGBoostClassifier 0.82377 100.00 % Jan 8, 2026 10:03 AM 30s booster
StandardScalerWrapper, XGBoostClassifier 0.81818 100.00 % Jan 8, 2026 10:03 AM 35s booster
SparseNormalizer, LightGBM 0.81814 100.00 % Jan 8, 2026 10:03 AM 35s boostin
MaxAbsScaler, XGBoostClassifier 0.81705 100.00 % Jan 8, 2026 10:03 AM 40s tree_me
MaxAbsScaler, LightGBM 0.81704 100.00 % Jan 8, 2026 10:03 AM 34s boostin
StandardScalerWrapper, RandomForest 0.81144 100.00 % Jan 8, 2026 10:03 AM 31s bootstra
MaxAbsScaler, LightGBM 0.80920 100.00 % Jan 8, 2026 10:03 AM 35s min_dat
StandardScalerWrapper, XGBoostClassifier 0.80355 100.00 % Jan 8, 2026 10:03 AM 36s booster
StandardScalerWrapper, XGBoostClassifier 0.80246 100.00 % Jan 8, 2026 10:03 AM 33s booster
Page 1 of 1 25/Page

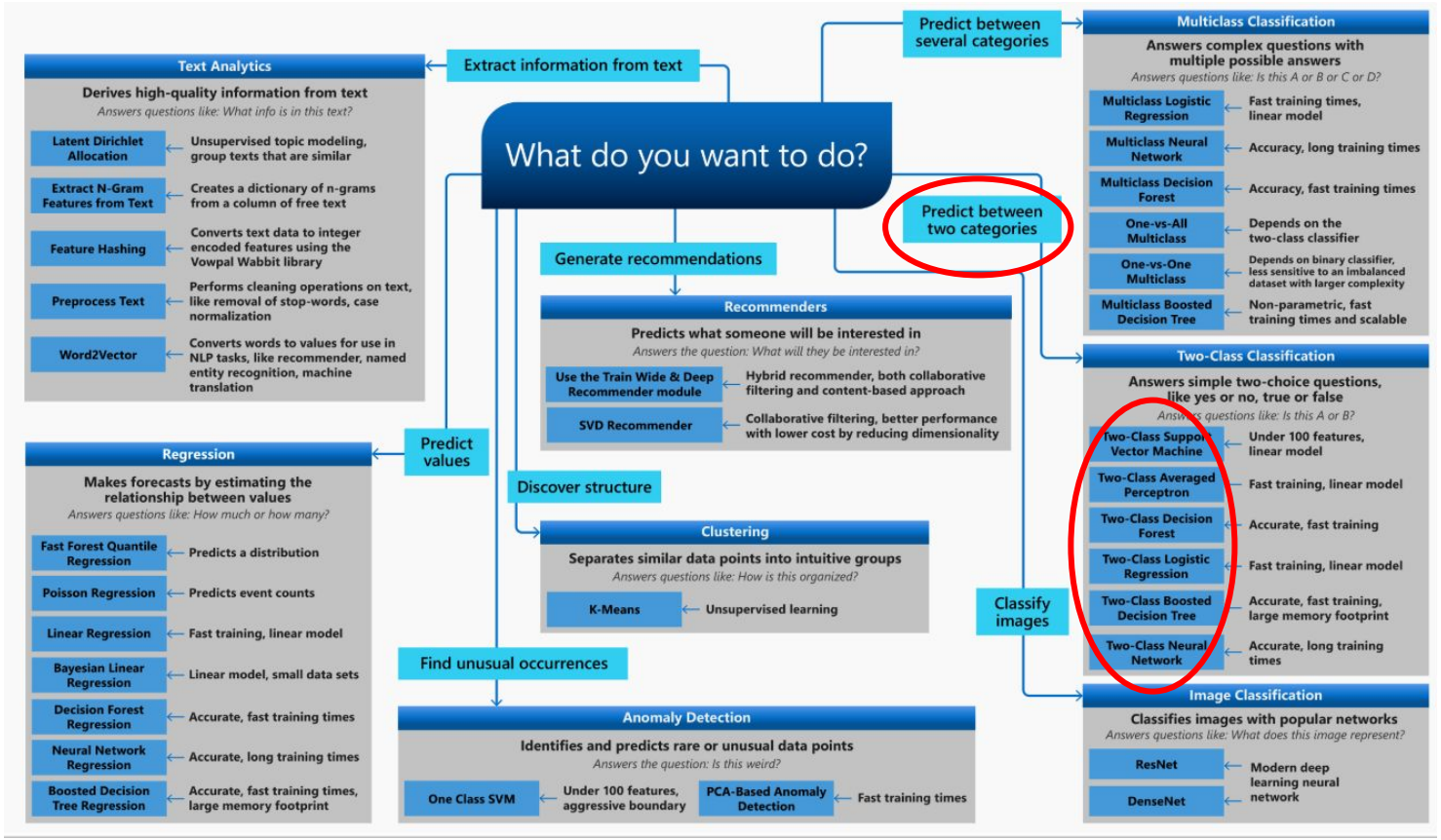
Notebooks: 실제로 코딩하면서 디테일 튜닝.

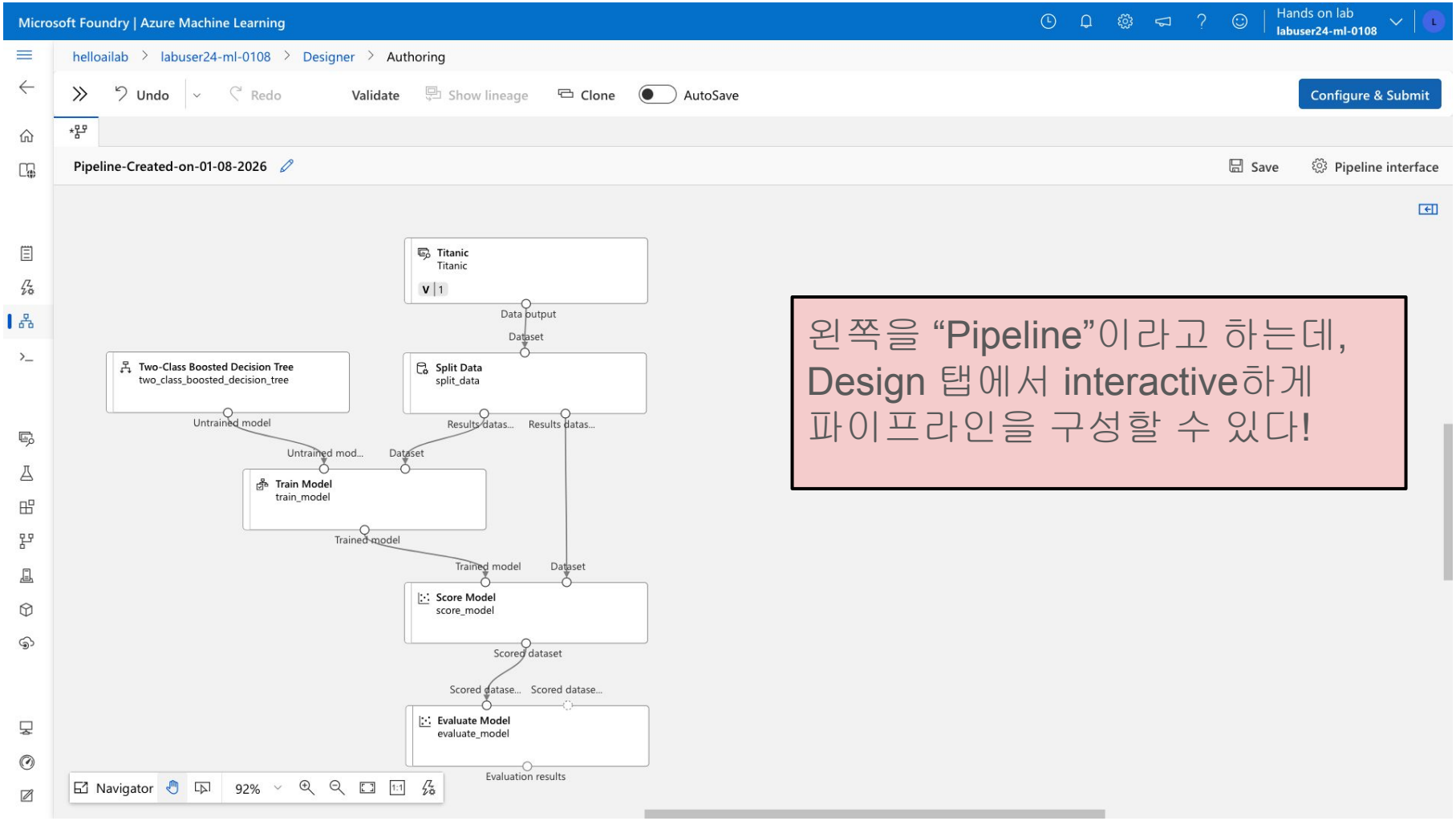


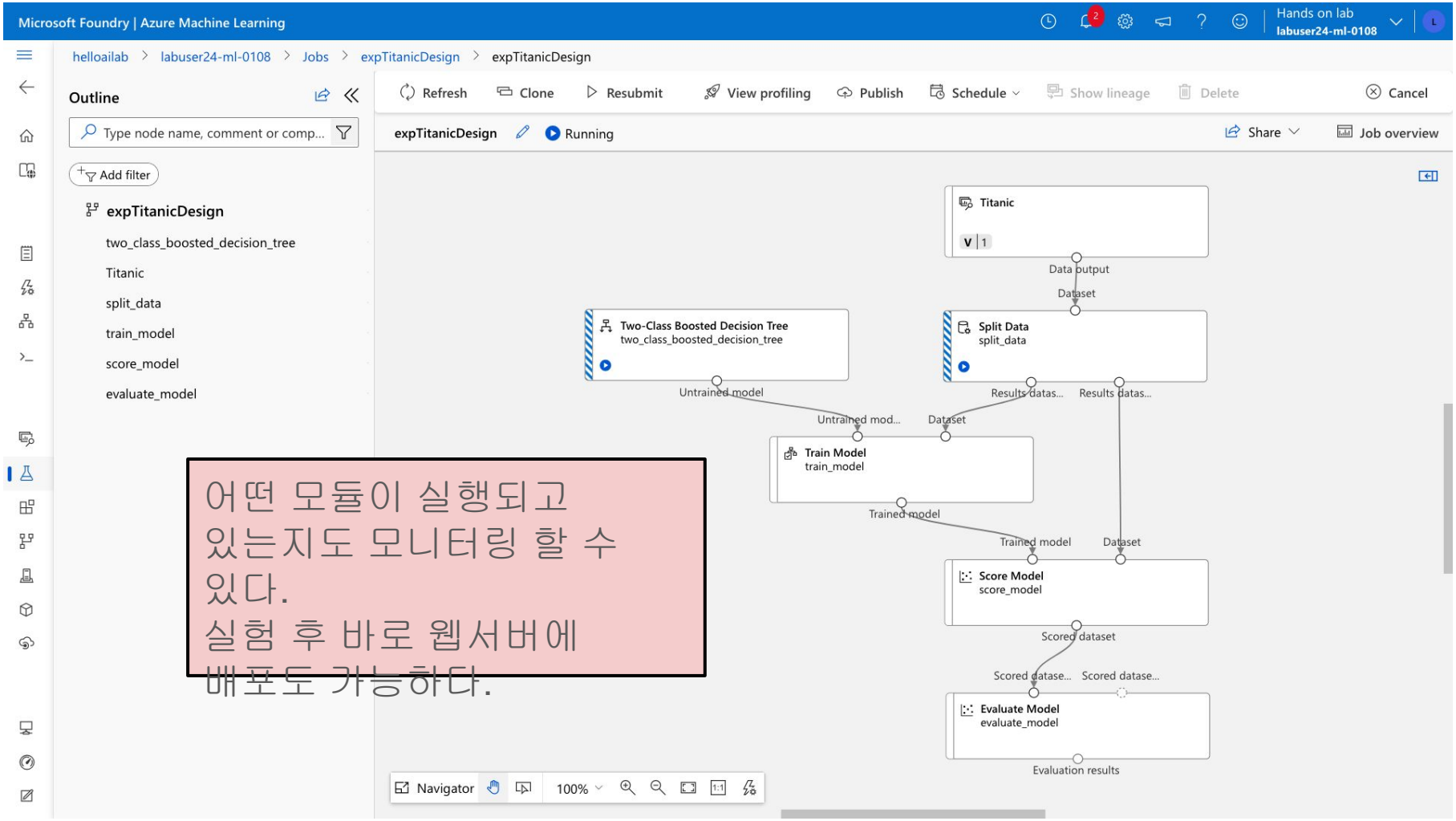

Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
helloailab
labuser24-ml-0108
Designer
Authoring
Undo
Redo
Validate
Show lineage
Clone
AutoSave
Configure & Submit
Search by name, tags and description
Tags : All
Add filter
Data
Component
95
Computer Vision (6)
Model Scoring & Evaluation (6)
Machine Learning Algorithms (19)
Text Analytics (7)
Python Language (2)
Data Input and Output (3)
Enter Data Manually
Microsoft
Enables entering and editing small datasets by typing
values. [Learn More](https://aka.ms/aml/enter-data-...
azureml.Designer:true
3/24/2025
Export Data
Microsoft
Writes a dataset to cloud-based storage in Azure, such
as Azure blob storage, Azure Data Lake Storage Gen...
azureml.Designer:true
5/8/2024
Pipeline-Created-on-01-08-2026
Save
Pipeline interface
Titanic
Titanic
V 1
Data output
Dataset
Split Data
split_data
Results datas...
Results datas...
Parameters
Interactive Coding?
Navigator
102%
Split Data
Splitting mode
Split Rows
Fraction of rows in the first output dataset
0.5
Randomized split
True
Random seed
0
Stratified split
False
Output settings
Input settings
Run settings
Node information
Component information


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
helloailab > labuser24-ml-0108 > Designer > Authoring
Undo
Redo
Validate
Show lineage
Clone
AutoSave
Configure & Submit
train
Tags : All
Add filter
Data
Component
39
Most relev...
Train Model
Microsoft
Trains a classification or regression model in a supervised manner. [Learn More](https://aka.ms...
azureml.Designer:true
3/24/2025
Train Clustering Model
Microsoft
Train clustering model and assign data to clusters. [Learn More](https://aka.ms/aml/train-...
azureml.Designer:true
3/24/2025
Train PyTorch Model
Microsoft
Train pytorch model from scratch or finetune it. [Learn More](https://aka.ms/aml/train-pytorch-...
azureml.Designer:true
12/21/2024
Train SVD Recommender
Microsoft
Train a collaborative filtering recommendation using SVD algorithm. [Learn More](https://aka....
azureml.Designer:true
3/24/2025
Pipeline-Created-on-01-08-2026
Save
Pipeline interface
EDA까지 이렇게
간편하게 해줌...
Two-Class Boosted Decision Tree
two_class_boosted_decision_tree
Untrained model
Untrained mod...
Dataset
Navigator
100%
DataOutput
Preview
Profile
Number of columns: 6
Number of rows: 891
Column
Profile
Sex
male
female
0
100
200
300
400
500
600
Count
Age
Count (approx)
100
0
0
20
40
60
Fare
500

cheet sheet


Machine Learning Algorithm Cheat Sheet
This cheat sheet helps you choose the best machine learning algorithm for your predictive analytics solution.
Your decision is driven by both the nature of your data and the goal you want to achieve with your data.
What do you want to do?
Extract information from text
Text Analytics
Derives high-quality information from text
Answers questions like: What info is in this text?
Latent Dirichlet Allocation
Unsupervised topic modeling, group texts that are similar
Extract N-Gram Features from Text
Creates a dictionary of n-grams from a column of free text
Feature Hashing
Converts text data to integer encoded features using the Vowpal Wabbit library
Preprocess Text
Performs cleaning operations on text, like removal of stop-words, case normalization
Word2Vector
Converts words to values for use in NLP tasks, like recommender, named entity recognition, machine translation
Predict between several categories
Multiclass Classification
Answers complex questions with multiple possible answers
Answers questions like: Is this A or B or C or D?
Multiclass Logistic Regression
Fast training times, linear model
Multiclass Neural Network
Accuracy, long training times
Multiclass Decision Forest
Accuracy, fast training times
One-vs-All Multiclass
Depends on the two-class classifier
One-vs-One Multiclass
Depends on binary classifier, less sensitive to an imbalanced dataset with larger complexity
Multiclass Boosted Decision Tree
Non-parametric, fast training times and scalable
Predict between two categories
Generate recommendations
Recommenders
Predicts what someone will be interested in
Answers the question: What will they be interested in?
Use the Train Wide & Deep Recommender module
Hybrid recommender, both collaborative filtering and content-based approach
SVD Recommender
Collaborative filtering, better performance with lower cost by reducing dimensionality
Two-Class Classification
Answers simple two-choice questions, like yes or no, true or false
Answers questions like: Is this A or B?
Two-Class Support Vector Machine
Under 100 features, linear model
Two-Class Averaged Perceptron
Fast training, linear model
Two-Class Decision Forest
Accurate, fast training
Two-Class Logistic Regression
Fast training, linear model
Two-Class Boosted Decision Tree
Accurate, fast training, large memory footprint
Two-Class Neural Network
Accurate, long training times
Predict values
Regression
Makes forecasts by estimating the relationship between values
Answers questions like: How much or how many?
Fast Forest Quantile Regression
Predicts a distribution
Poisson Regression
Predicts event counts
Linear Regression
Fast training, linear model
Bayesian Linear Regression
Linear model, small data sets
Decision Forest Regression
Accurate, fast training times
Neural Network Regression
Accurate, long training times
Boosted Decision Tree Regression
Accurate, fast training times, large memory footprint
Discover structure
Clustering
Separates similar data points into intuitive groups
Answers questions like: How is this organized?
K-Means
Unsupervised learning
Classify images
Image Classification
Classifies images with popular networks
Answers questions like: What does this image represent?
ResNet
Modern deep learning neural network
DenseNet
Find unusual occurrences
Anomaly Detection
Identifies and predicts rare or unusual data points
Answers the question: Is this weird?
One Class SVM
Under 100 features, aggressive boundary
PCA-Based Anomaly Detection
Fast training times


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
helloailab > labuser24-ml-0108 > Designer > Authoring
Undo
Redo
Validate
Show lineage
Clone
AutoSave
Configure & Submit
Pipeline-Created-on-01-08-2026
Save
Pipeline interface
Titanic
Titanic
v 1
Data output
Dataset
Two-Class Boosted Decision Tree
two_class_boosted_decision_tree
Split Data
split_data
Untrained model
Results datas...
Results datas...
Untrained mod...
Dataset
Train Model
train_model
Trained model
Trained model
Dataset
Score Model
score_model
Scored dataset
Scored datase...
Scored datase...
Evaluate Model
evaluate_model
Evaluation results
Navigator
92%
왼쪽을 “Pipeline”이라고 하는데,
Design 탭에서 interactive하게
파이프라인을 구성할 수 있다!


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
helloailab > labuser24-ml-0108 > Jobs > expTitanicDesign > expTitanicDesign
Outline
Type node name, comment or comp...
Add filter
expTitanicDesign
two_class_boosted_decision_tree
Titanic
split_data
train_model
score_model
evaluate_model
Refresh
Clone
Resubmit
View profiling
Publish
Schedule
Show lineage
Delete
Cancel
expTitanicDesign
Running
Share
Job overview
Titanic
v 1
Data output
Dataset
Two-Class Boosted Decision Tree
two_class_boosted_decision_tree
Split Data
split_data
Untrained model
Results datas...
Results datas...
Untrained mod...
Dataset
Train Model
train_model
Trained model
Trained model
Dataset
Score Model
score_model
Scored dataset
Scored datase...
Scored datase...
Evaluate Model
evaluate_model
Evaluation results
Navigator
100%
어떤 모듈이 실행되고 있는지도 모니터링 할 수 있다.
실험 후 바로 웹서버에 배포도 가능하다.

Publish succeeded. Go to pipeline endpoint Pipeline-Created-on-01-08-2026

Outline

Type node name, comment or comp...

Add filter

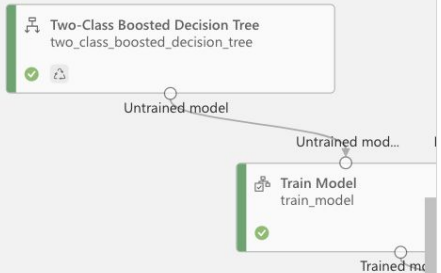
Pipeline-Created-on-01-08-2026

- two_class_boosted_decision_tree
- Titanic
- split_data
- train_model
- score_model
- evaluate_model

Refresh | Clone | Resubmit | View profiling | Publish | Schedule | Show lineage | ... | Cancel

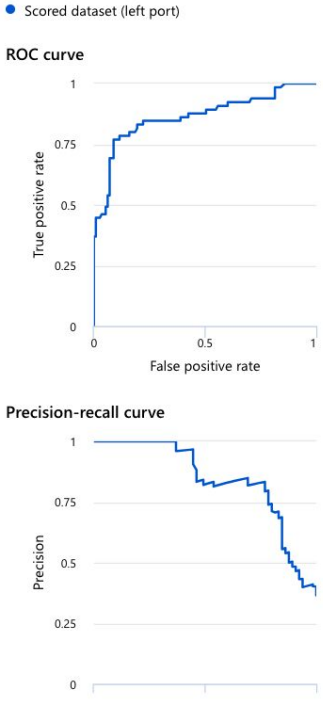
Pipeline-Created-on-01-08-2026 Completed

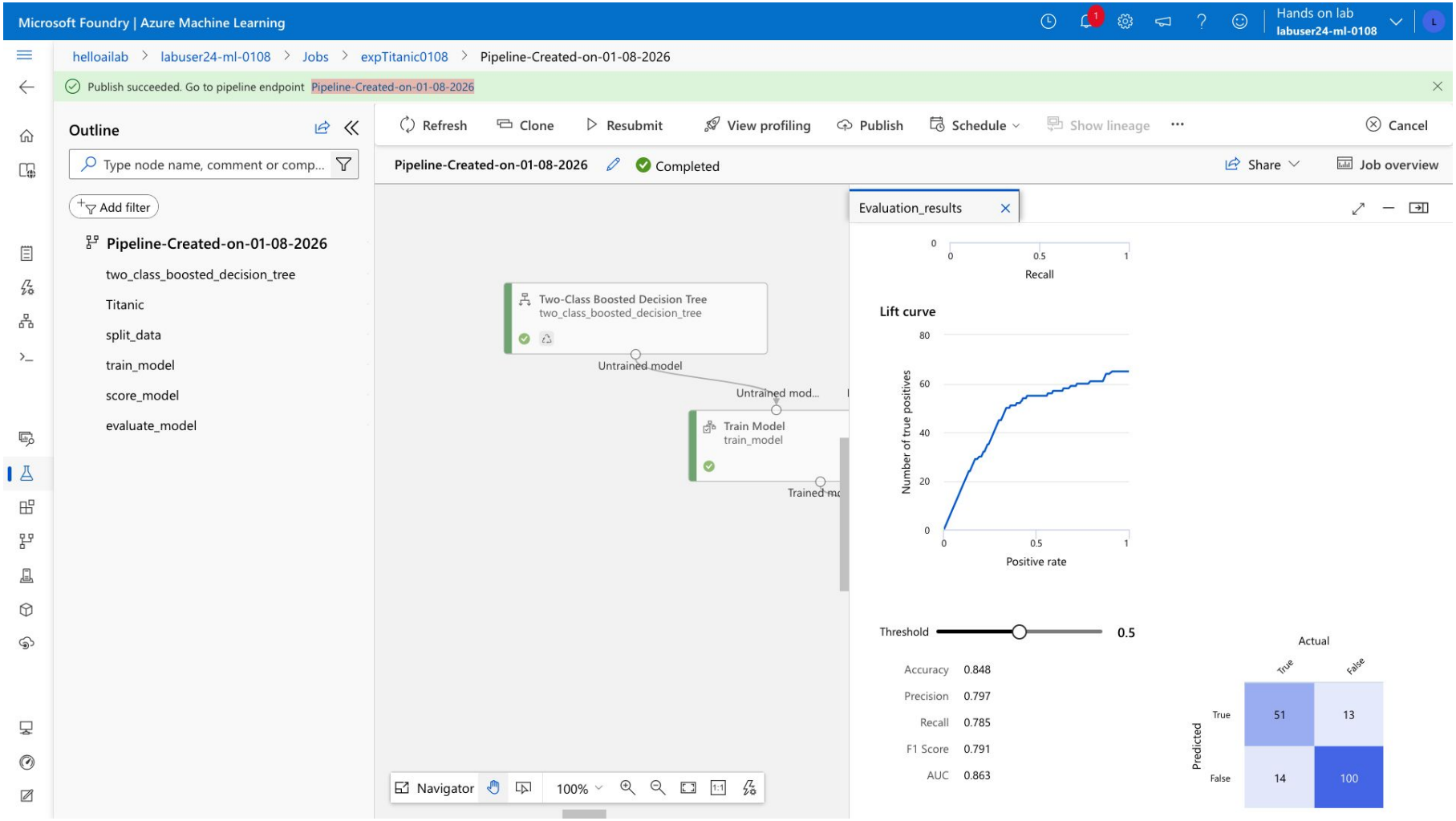
Share | Job overview



Navigator | 100%

Evaluation_results




Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
helloailab > labuser24-ml-0108 > Jobs > expTitanic0108 > Pipeline-Created-on-01-08-2026
Publish succeeded. Go to pipeline endpoint Pipeline-Created-on-01-08-2026
Outline
Type node name, comment or comp...
Add filter
Pipeline-Created-on-01-08-2026
two_class_boosted_decision_tree
Titanic
split_data
train_model
score_model
evaluate_model
Refresh
Clone
Resubmit
View profiling
Publish
Schedule
Show lineage
Cancel
Pipeline-Created-on-01-08-2026
Completed
Share
Job overview
Two-Class Boosted Decision Tree
two_class_boosted_decision_tree
Untrained model
Untrained mod...
Train Model
train_model
Trained mo
Navigator
100%
Evaluation_results
Recall
Lift curve
Number of true positives
Positive rate
Threshold
0.5
Accuracy 0.848
Precision 0.797
Recall 0.785
F1 Score 0.791
AUC 0.863
Actual
True
False
Predicted
51
13
14
100

Pipeline-Created-on-01-08-2026

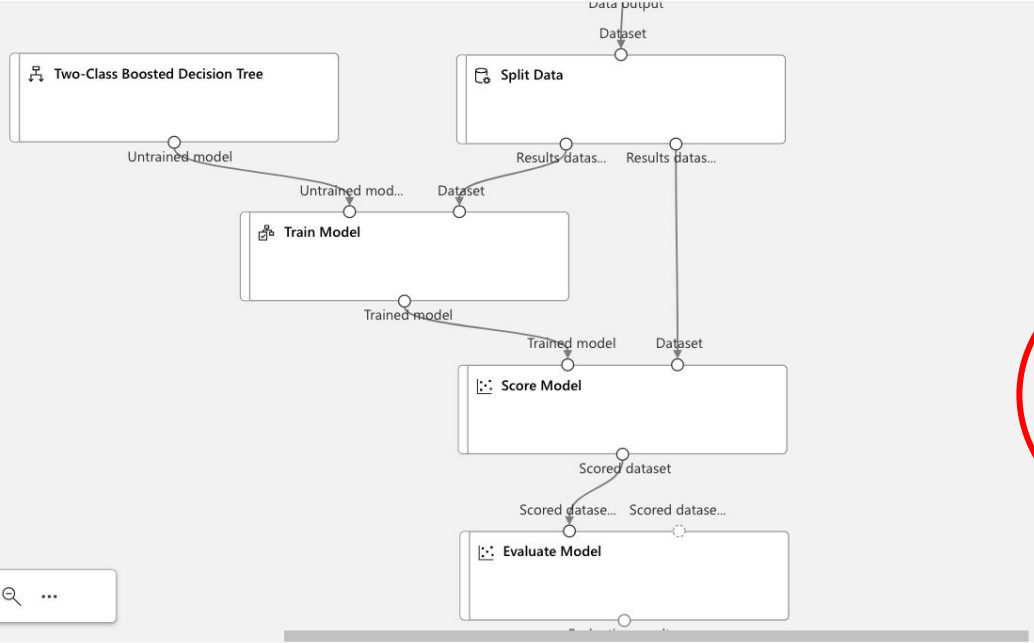
배포(publish) 후

Details | Published pipelines

Refresh | Submit | Clone | Show lineage

Search canvas

Hide overview



Navigator 100%

Pipeline endpoint overview

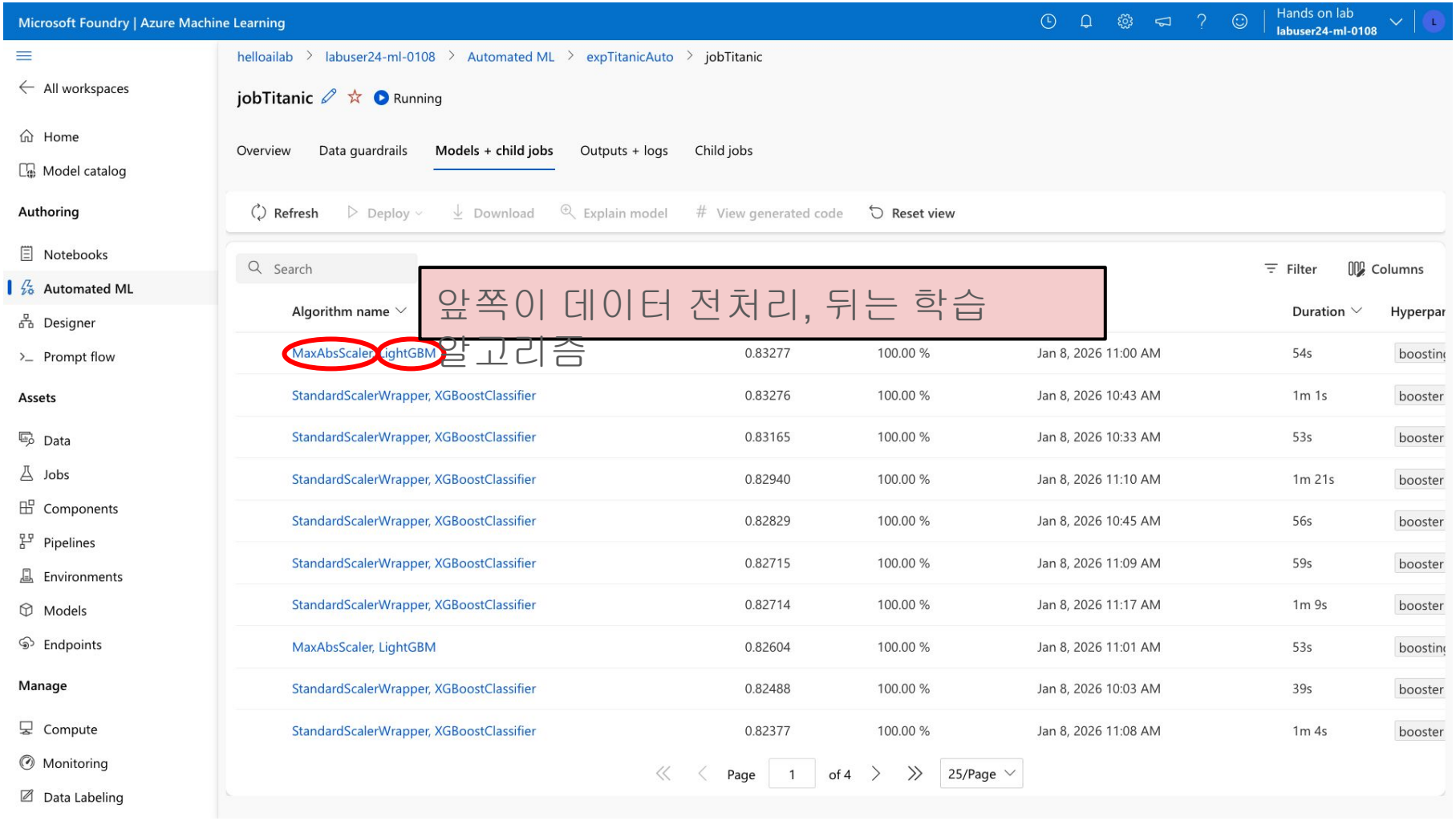
Overview | Settings

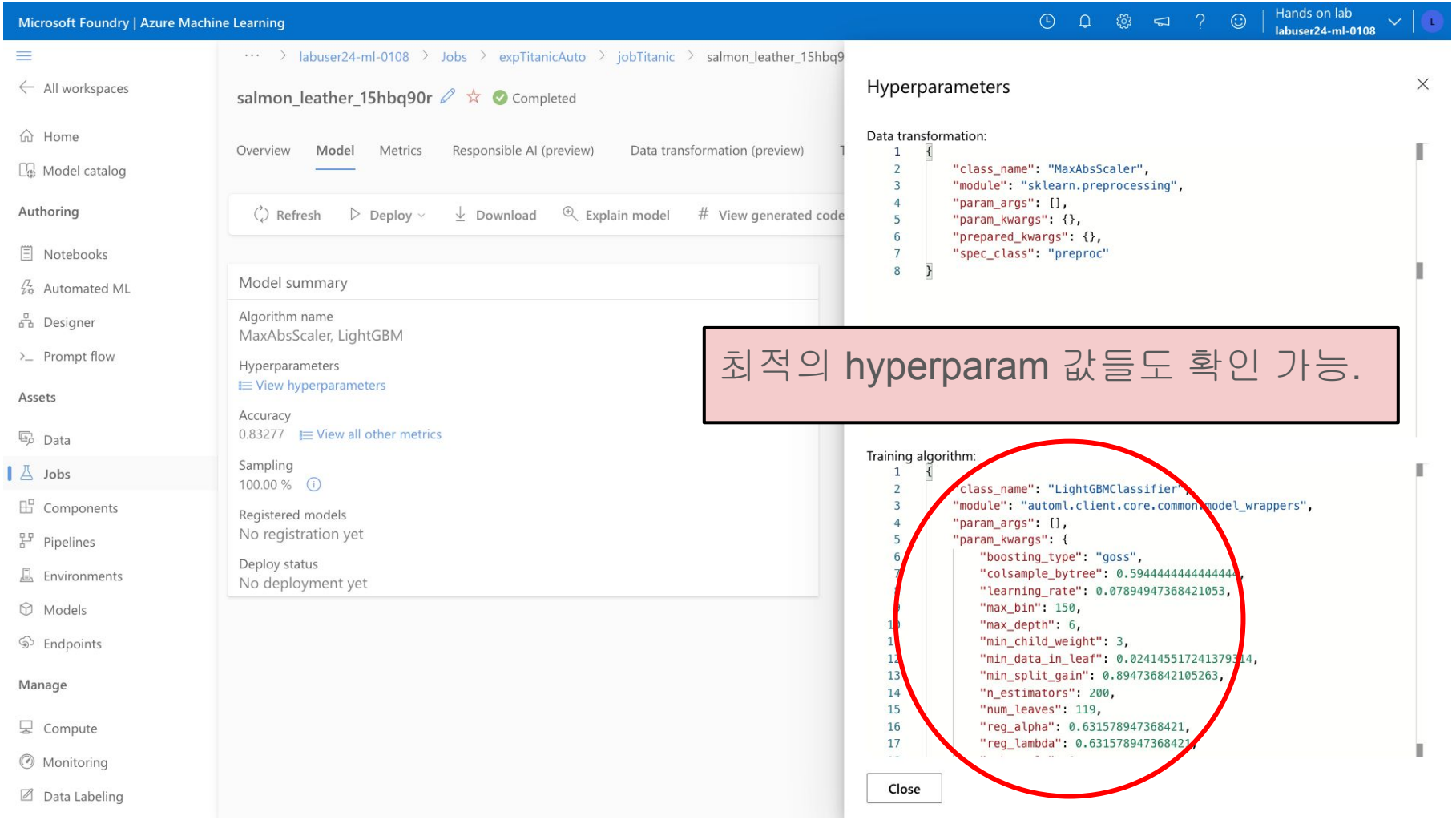
Properties

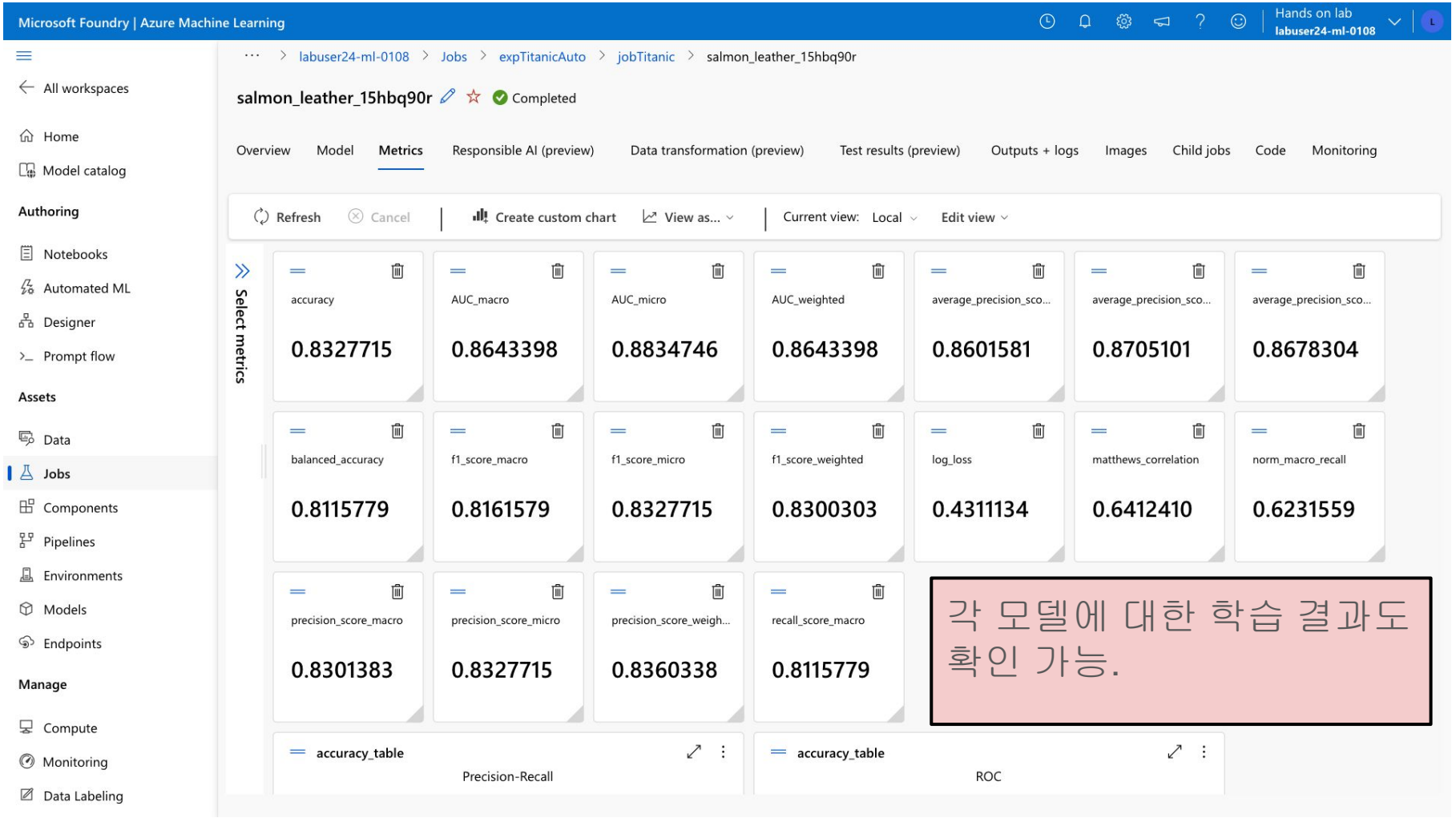
Status
Active

REST endpoint
https://spaincentral.api.azureml.ms/pipelines/v1.0/subscriptions/14734032-3a81-427c-b9a7-6a15145822d4/resourceGroups/rg-labuser24/providers/Microsoft.MachineLearningServices/workspaces/labuser24-ml-0108/PipelineRuns/PipelineEndpointSubmit/Id/c35624f9-faa6-41a1-a4eb-0045fb368c4c

REST endpoint documentation
https://spaincentral.api.azureml.ms/pipelines/swagger/pipelineendpointsubmit/swagger.json


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
helloailab > labuser24-ml-0108 > Automated ML > expTitanicAuto > jobTitanic
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
jobTitanic
Running
Overview
Data guardrails
Models + child jobs
Outputs + logs
Child jobs
Refresh
Deploy
Download
Explain model
View generated code
Reset view
Search
Filter
Columns
앞쪽이 데이터 전처리, 뒤는 학습 알고리즘
Algorithm name
Duration
Hyperpar
MaxAbsScaler, LightGBM 0.83277 100.00 % Jan 8, 2026 11:00 AM 54s boostin
StandardScalerWrapper, XGBoostClassifier 0.83276 100.00 % Jan 8, 2026 10:43 AM 1m 1s booster
StandardScalerWrapper, XGBoostClassifier 0.83165 100.00 % Jan 8, 2026 10:33 AM 53s booster
StandardScalerWrapper, XGBoostClassifier 0.82940 100.00 % Jan 8, 2026 11:10 AM 1m 21s booster
StandardScalerWrapper, XGBoostClassifier 0.82829 100.00 % Jan 8, 2026 10:45 AM 56s booster
StandardScalerWrapper, XGBoostClassifier 0.82715 100.00 % Jan 8, 2026 11:09 AM 59s booster
StandardScalerWrapper, XGBoostClassifier 0.82714 100.00 % Jan 8, 2026 11:17 AM 1m 9s booster
MaxAbsScaler, LightGBM 0.82604 100.00 % Jan 8, 2026 11:01 AM 53s boostin
StandardScalerWrapper, XGBoostClassifier 0.82488 100.00 % Jan 8, 2026 10:03 AM 39s booster
StandardScalerWrapper, XGBoostClassifier 0.82377 100.00 % Jan 8, 2026 11:08 AM 1m 4s booster
Page 1 of 4
25/Page


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
labuser24-ml-0108 > Jobs > expTitanicAuto > jobTitanic > salmon_leather_15hbq9
salmon_leather_15hbq90r
Completed
Overview
Model
Metrics
Responsible AI (preview)
Data transformation (preview)
Refresh
Deploy
Download
Explain model
View generated code
Model summary
Algorithm name
MaxAbsScaler, LightGBM
Hyperparameters
View hyperparameters
Accuracy
0.83277
View all other metrics
Sampling
100.00 %
Registered models
No registration yet
Deploy status
No deployment yet
Hyperparameters
Data transformation:
{
"class_name": "MaxAbsScaler",
"module": "sklearn.preprocessing",
"param_args": [],
"param_kwargs": {},
"prepared_kwargs": {},
"spec_class": "preproc"
}
최적의 hyperparam 값들도 확인 가능.
Training algorithm:
{
"class_name": "LightGBMClassifier",
"module": "automl.client.core.common.model_wrappers",
"param_args": [],
"param_kwargs": {
"boosting_type": "goss",
"colsample_bytree": 0.5944444444444444,
"learning_rate": 0.07894947368421053,
"max_bin": 150,
"max_depth": 6,
"min_child_weight": 3,
"min_data_in_leaf": 0.024145517241379314,
"min_split_gain": 0.894736842105263,
"n_estimators": 200,
"num_leaves": 119,
"reg_alpha": 0.631578947368421,
"reg_lambda": 0.631578947368421,
Close


Microsoft Foundry | Azure Machine Learning
Hands on lab
labuser24-ml-0108
All workspaces
Home
Model catalog
Authoring
Notebooks
Automated ML
Designer
Prompt flow
Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints
Manage
Compute
Monitoring
Data Labeling
labuser24-ml-0108 > Jobs > expTitanicAuto > jobTitanic > salmon_leather_15hbq90r
salmon_leather_15hbq90r
Completed
Overview
Model
Metrics
Responsible AI (preview)
Data transformation (preview)
Test results (preview)
Outputs + logs
Images
Child jobs
Code
Monitoring
Refresh
Cancel
Create custom chart
View as...
Current view: Local
Edit view
Select metrics
accuracy 0.8327715
AUC_macro 0.8643398
AUC_micro 0.8834746
AUC_weighted 0.8643398
average_precision_sco... 0.8601581
average_precision_sco... 0.8705101
average_precision_sco... 0.8678304
balanced_accuracy 0.8115779
f1_score_macro 0.8161579
f1_score_micro 0.8327715
f1_score_weighted 0.8300303
log_loss 0.4311134
matthews_correlation 0.6412410
norm_macro_recall 0.6231559
precision_score_macro 0.8301383
precision_score_micro 0.8327715
precision_score_weigh... 0.8360338
recall_score_macro 0.8115779
각 모델에 대한 학습 결과도 확인 가능.
accuracy_table
Precision-Recall
accuracy_table
ROC

All workspaces

Home
Model catalog

Authoring
Notebooks
Automated ML
Designer
Prompt flow

Assets
Data
Jobs
Components
Pipelines
Environments
Models
Endpoints

Manage
Compute
Monitoring
Data Labeling

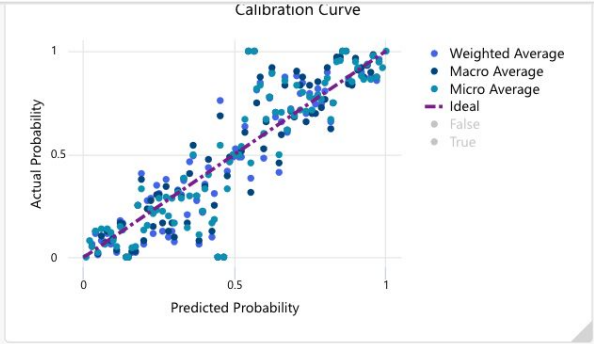
salmon_leather_15hbq90r

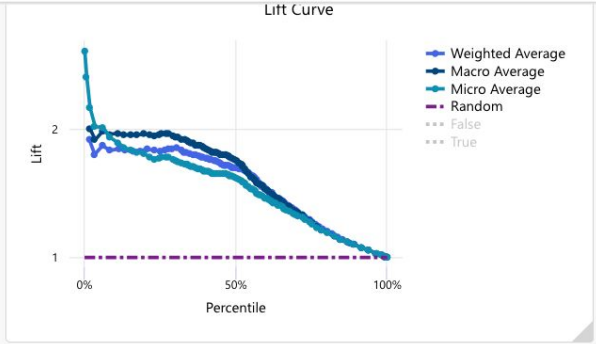
Completed

Overview | Model | Metrics | Responsible AI (preview) | Data transformation (preview) | Test results (preview) | Outputs + logs | Images | Child jobs | Code | Monitoring

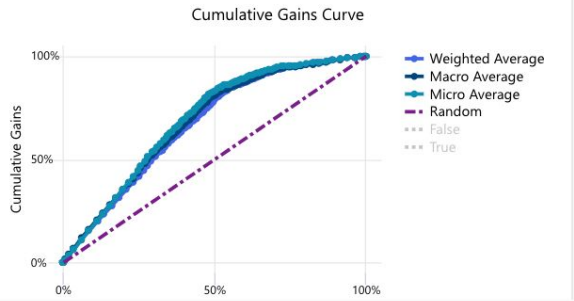
Refresh | Cancel | Create custom chart | View as... | Current view: Local | Edit view

Select metrics

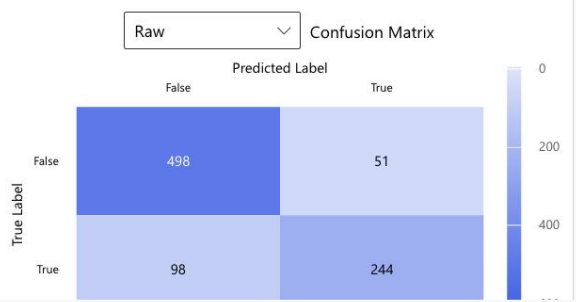


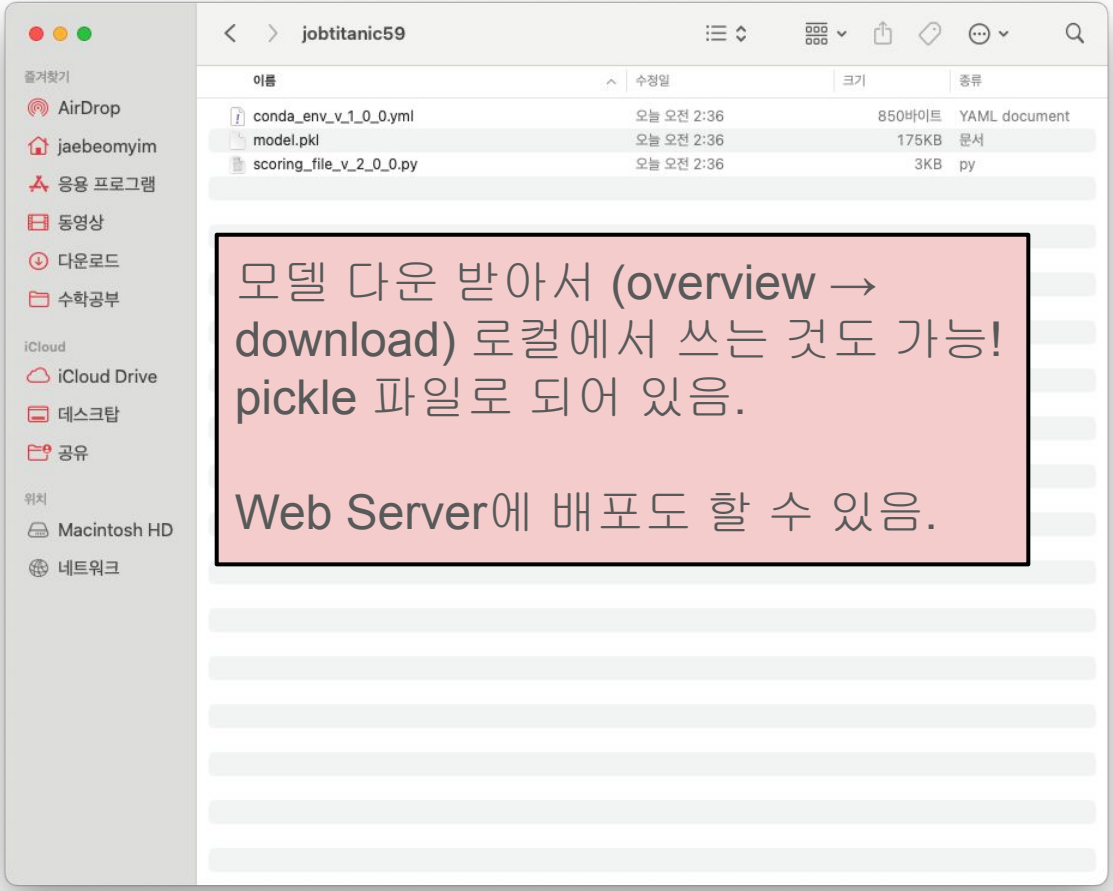


accuracy_table



confusion_matrix




jobtitanic59
즐겨찾기
AirDrop
jaebeomyim
응용 프로그램
동영상
다운로드
수학공부
iCloud
iCloud Drive
데스크탑
공유
위치
Macintosh HD
네트워크
이름
수정일
크기
종류
conda_env_v_1_0_0.yml
오늘 오전 2:36
850바이트
YAML document
model.pkl
오늘 오전 2:36
175KB
문서
scoring_file_v_2_0_0.py
오늘 오전 2:36
3KB
py
모델 다운 받아서 (overview → download) 로컬에서 쓰는 것도 가능!
pickle 파일로 되어 있음.
Web Server에 배포도 할 수 있음.

Microsoft Foundry(ai.azure.com)에서 보다 다양한 LLM 모델 사용하기

(11:30 ~) Hugging Face

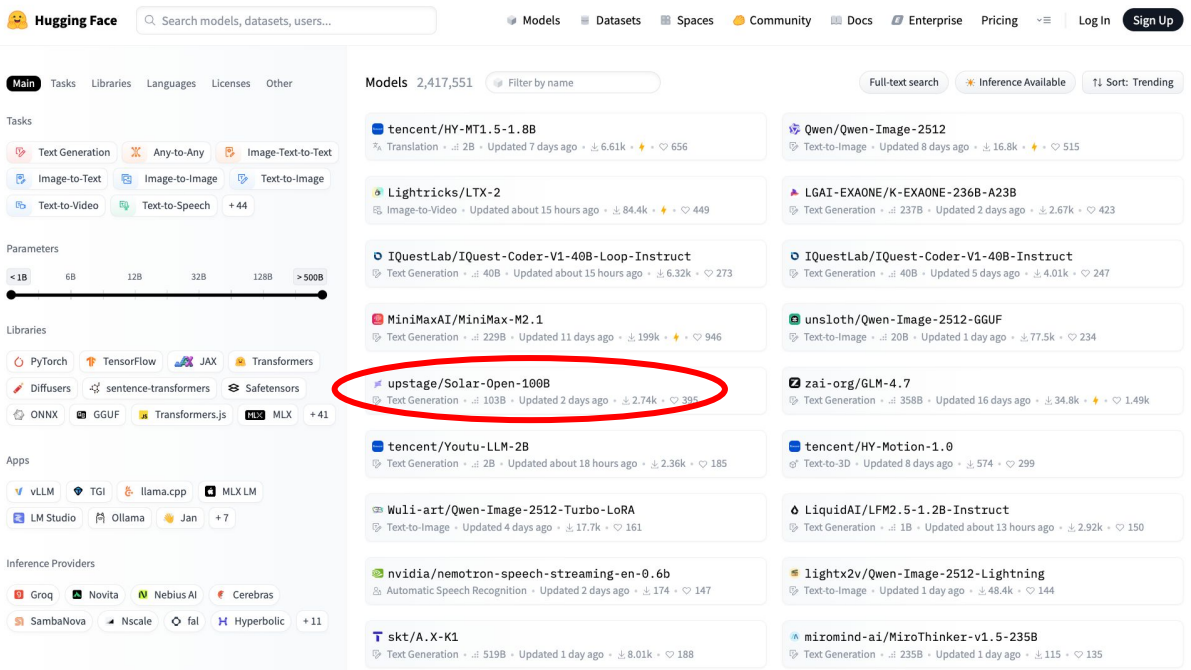
AI 모델계의 github 같은 곳.

https://huggingface.co/

ChatGPT 만 해도 충분한데, 왜 이렇게 모델들이 많이 나와있을까? (upstage는 한국 기업: 업스테이지)

다양한 용도와 수요들이 전부 커버가 안됨…

Huggingface에 올라와 있는 모델들을 사용자가 Azure cloud에서 pipeline에 삽입하여 사용하고, 사용자가 Azure에 지불하면 Azure가 HF에 비용 지불하고… 이런 지불구조.



Microsoft Foundry

New Foundry | Docs

Keep building with Microsoft Foundry

View all resources | + Create new | Help

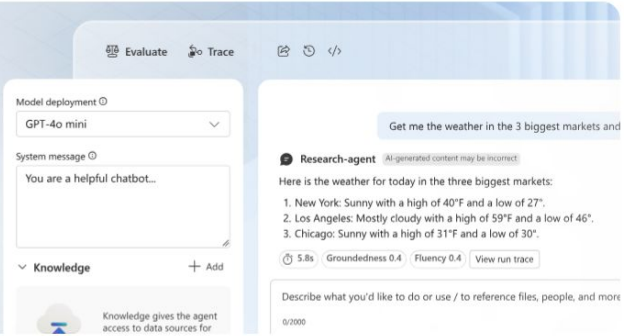
Resource name	Type	Region	Created on
labuser11-0108	Project (default) labuser11-0108-resource (AI Foundry)	eastus2	

Explore models and capabilities

Create smarter agents with Microsoft Foundry

Design, customize, and manage powerful, adaptable AI agents that automate tasks, integrate seamlessly with your apps, and enhance user experiences.

Create an agent



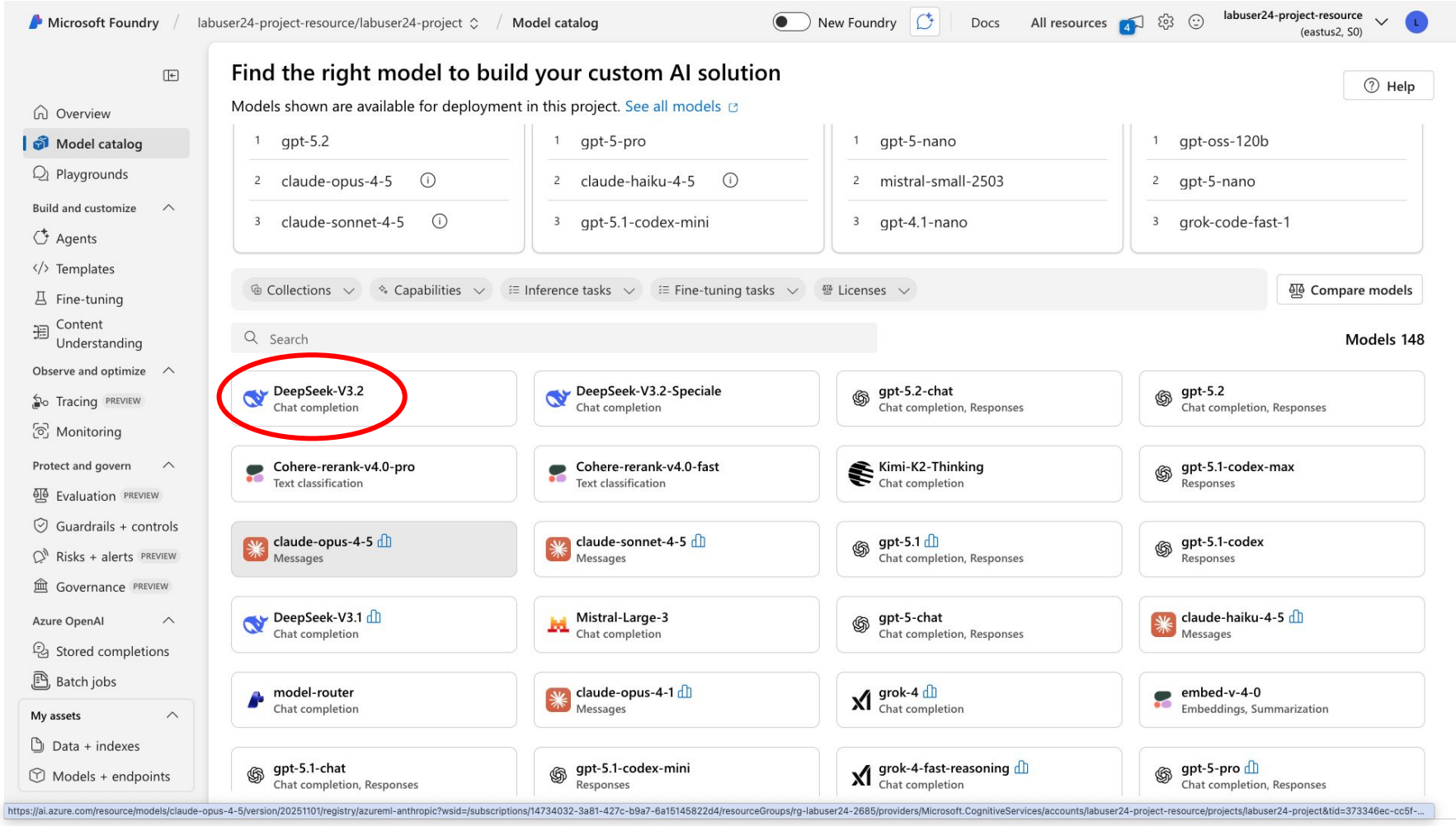
Search for a model

Latest models

Go to full model catalog

- DeepSeek-V3.2 — Chat completion
- DeepSeek-V3.2-Speciale — Chat completion
- gpt-5.2-chat — Chat completion, Responses

https://ai.azure.com/?tid=373346ec-cc5f-4210-b5bc-9a4c2d9517d8&doNotRedirectLastResource=true



https://ai.azure.com/resource/models/claude-opus-4-5/version/20251101/registry/azureml-anthropic?wsid=/subscriptions/14734032-3a81-427c-b9a7-6a15145822d4/resourceGroups/rg-labuser24-2685/providers/Microsoft.CognitiveServices/accounts/labuser24-project-resource/projects/labuser24-project&tid=373346ec-cc5f-...

Overview
Model catalog
Playgrounds
Build and customize
Agents
Templates
Fine-tuning
Content Understanding
Observe and optimize
Tracing PREVIEW
Monitoring
Protect and govern
Evaluation PREVIEW
Guardrails + controls
Risks + alerts PREVIEW
Governance PREVIEW
Azure OpenAI
Stored completions
Batch jobs
My assets
Data + indexes
Models + endpoints

DeepSeek-V3.2-2

Help

Details | Metrics

Open in playground | Request quota | Edit | Delete

Endpoint

Target URI

https://labuser16-project-resource.services.ai.azure.com/m...

Key

••...

Deployment info

Name DeepSeek-V3.2-2	**Provisioning state** Succeeded
Deployment type Global Standard	**Created on** 2026-01-08T03:00:43.2985998Z
Created by labuser24@helloaicloud.onmicrosoft.com	**Modified on** Jan 8, 2026 12:00 PM
Modified by labuser24@helloaicloud.onmicrosoft.com	**Version upgrade policy** Once a new default version is available
Rate limit (Tokens per minute) 250,000	**Rate limit (Requests per minute)** 250
Model name DeepSeek-V3.2	**Model version** 1
Life cycle status	**Date created**

Language: Python | **SDK:** OpenAI SDK | **Authentication type:** Key Authentication

Get Started

Below are example code snippets for a few use cases. For additional information about OpenAI SDK, see full documentation and samples.

1. Authentication using API Key

For OpenAI API Endpoints, deploy the Model to generate the endpoint URL and an API key to authenticate against the service. In this sample endpoint and key are strings holding the endpoint URL and the API Key.

The API endpoint URL and API key can be found on the Deployments + Endpoint page once the model is deployed.

To create a client with the OpenAI SDK using an API key, initialize the client by passing your API key to the SDK's configuration. This allows you to authenticate and interact with OpenAI's services seamlessly:

```
from openai import OpenAI

client = OpenAI(
    base_url=f"{endpoint}",
    api_key=api_key
)
```

2. Install dependencies

Install the Open AI SDK using pip (Requires: Python >=3.8):

```
pip install openai
```

Flows are here! Build and orchestrate smarter multi-agent workflows with speed and control. *Explore Flows*

crewai Home Enterprise Open Source Ecosystem Case Studies Use Cases Webinars Blog Sign in Start Cloud Trial

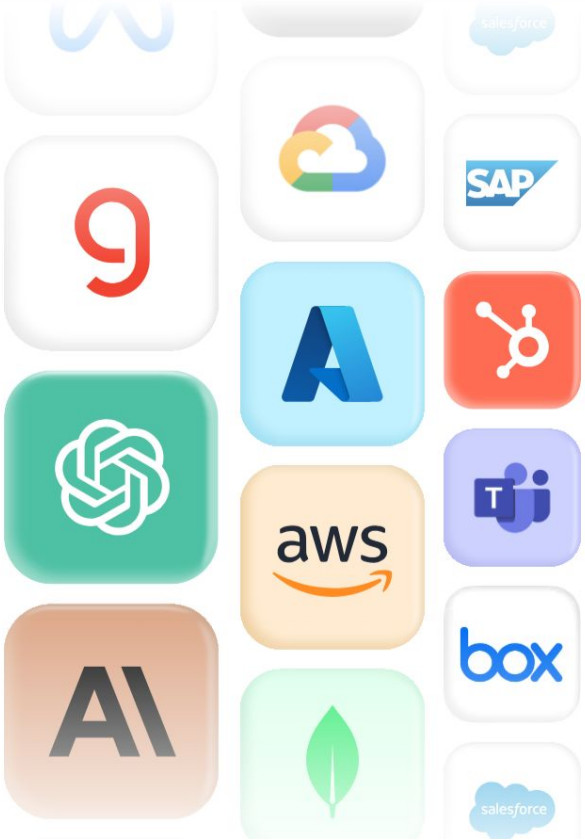
The Leading Multi-Agent Platform

Streamline workflows across industries with powerful AI agents. Build and deploy automated workflows using any LLM and cloud platform.

Request a Demo

500,000,000+

Multi-Agent Crews run using CrewAI



crewai: LangChain 처럼 에이전트 프레임워크 LangChain은 코드가 좀 지저분한 면이 있어서

Find by title

- Agent Framework
- Quick Start Guide
- Tutorials
- User Guide
- Integrations
- Support
- Migration Guide
- API Reference Guide
  - .NET API reference
  - Python API reference

microsoft에서 만든 LangChain 대용 에이전트 프레임워크

Download PDF

Microsoft Agent Framework

Microsoft Agent Framework is an open-source development kit for building **AI agents** and **multi-agent workflows** for .NET and Python. It brings together and extends ideas from Semantic Kernel and AutoGen projects, combining their strengths while adding new capabilities. Built by the same teams, it is the unified foundation for building AI agents going forward.

Agent Framework offers two primary categories of capabilities:

- AI agents: Individual agents that use LLMs to process user inputs, call tools and MCP servers to perform actions, and generate responses. Agents support model providers including Azure OpenAI, OpenAI, and Azure AI.
- Workflows: Graph-based workflows that connect multiple agents and functions to perform complex, multi-step tasks. Workflows support type-based routing, nesting, checkpointing, and request/response patterns for human-in-the-loop scenarios.

The framework also provides foundational building blocks, including model clients (chat completions and responses), an agent thread for state management, context providers for agent memory, middleware for intercepting agent actions, and MCP clients for tool integration. Together, these components give you the flexibility and power to build interactive, robust, and safe AI applications.

Why another agent framework?

Semantic Kernel and AutoGen pioneered the concepts of AI agents and multi-agent orchestration. The Agent Framework is the direct successor, created by the same teams. It combines AutoGen's simple abstractions for single- and multi-agent patterns with Semantic Kernel's enterprise-grade features such as thread-based state management, type safety, filters, telemetry, and extensive

In this article

- Why another agent framework?
- Installation
- AI Agents
- Workflows
- Next steps

Was this page helpful?

Yes | No

그래서 LangChain이 뭐라고?

랭체인(LangChain): 대규모 언어 모델을 활용한 혁신적인 프레임워크

랭체인(LangChain)은 해리슨 체이스(Harrison Chase)에 의해 2022년 10월에 오픈 소스 프로젝트로 시작되었습니다. 그는 머신러닝 스타트업인 로버스트 인텔리전스(Robust Intelligence)에서 근무하면서 대규모 언어 모델(LLM)을 활용하여 애플리케이션과 파이프라인을 신속하게 구축할 수 있는 플랫폼의 필요성을 느꼈습니다. 이러한 비전을 가지고 개발자들이 챗봇, 질의응답 시스템, 자동 요약 등 다양한 LLM 애플리케이션을 쉽게 개발할 수 있도록 지원하는 프레임워크를 만들었습니다.

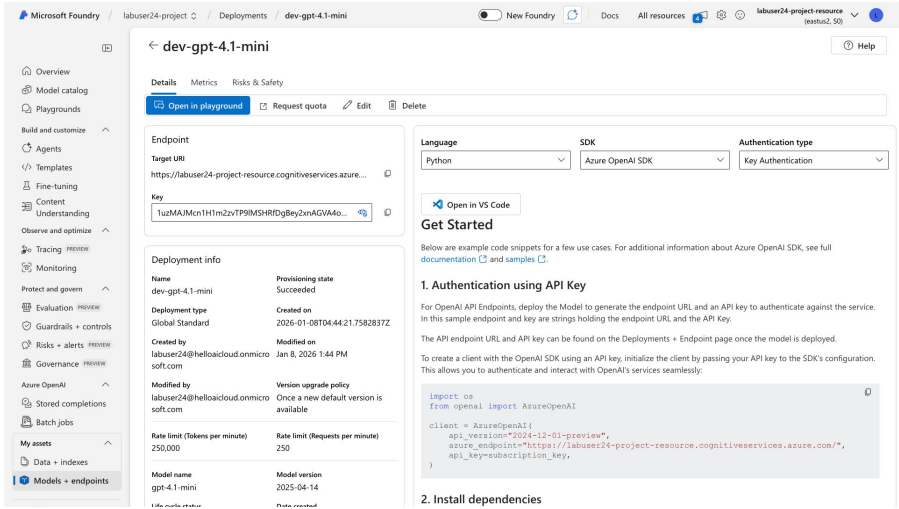
2023년 4월, 랭체인은 법인으로 전환하고 세쿼이아캐피털 등 벤처캐피탈의 투자를 받으며 빠르게 성장하고 있습니다. 2024년 1월, 최초의 안정적(stable) 버전인 v0.1.0을 공개하고 개발자 친화적인 LLM 개발 프레임워크로 위상을 강화하고 있습니다. 이전 버전까지는 잦은 업데이트로 인한 버그와 코드 오류 등으로 종종 개발자들을 불편하게 하기도 했지만, 새로운 버전이 출시되면서 이런 문제도 상당 부분 개선될 것으로 보입니다.

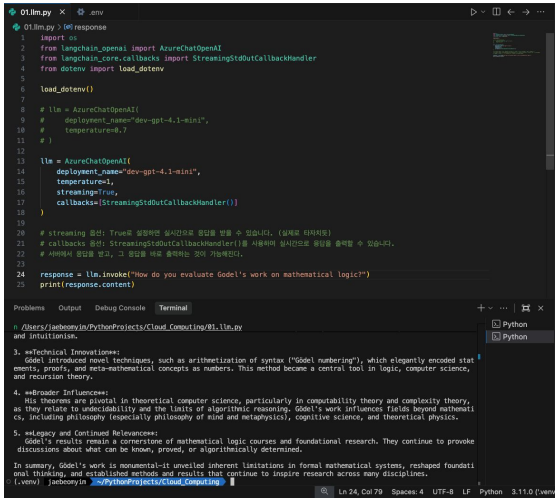
Afternoon Session

1. ai.azure.com (Microsoft Foundry)에서 직접 모델을 선택(deepseek, gpt 4.1 mini, Claude, etc.)하고 api key 값과 endpoint url을 생성한다.

2. local에서 langchain 라이브러리를 사용용해서 (advantage: it serves as a class wrapper, so the syntax is consistent with other llm's!) 선택한 모델 호출.

3. 늘 하던대로 llm.invoke()하면 되는데, streaming=True와 StreamingStdOutCallbackHandler() 설정하면 실시간 응답 가능.





결국 takeaway lesson은, langchain을 활용하면 어떤 LLM을 사용하든지 동일한 문법으로 동일한 task를 수행할 수 있다는 것!

+ (김상엽): 그것만 하고 싶으면 그냥 너가 class wrapper 하나 만들어서 쓰면 되잖아? Langchain을 쓰면 (그 문법의 통일성 덕분에…?) llm을 여러개 호출해서 (e.g. RAG → gpt → …) 파이프라인을 구축할 수 있다.

Langchain 쓰기 귀찮다? [illegible]
딸깍~

요약하자면

사용자님의 이해(API 통일성, 파이프라인)는 아주 정확합니다. 거기에 다음 두 가지를 더하면 완벽합니다.

1. **데이터 연결 (Data Connection):** 외부 데이터(PDF, DB)를 LLM과 쉽게 연결해준다.
2. **에이전트 (Agents):** LLM이 단순히 텍스트만 뱉는 게 아니라, 검색도 하고 코딩도 하고 계산기도 두드리게 만든다.

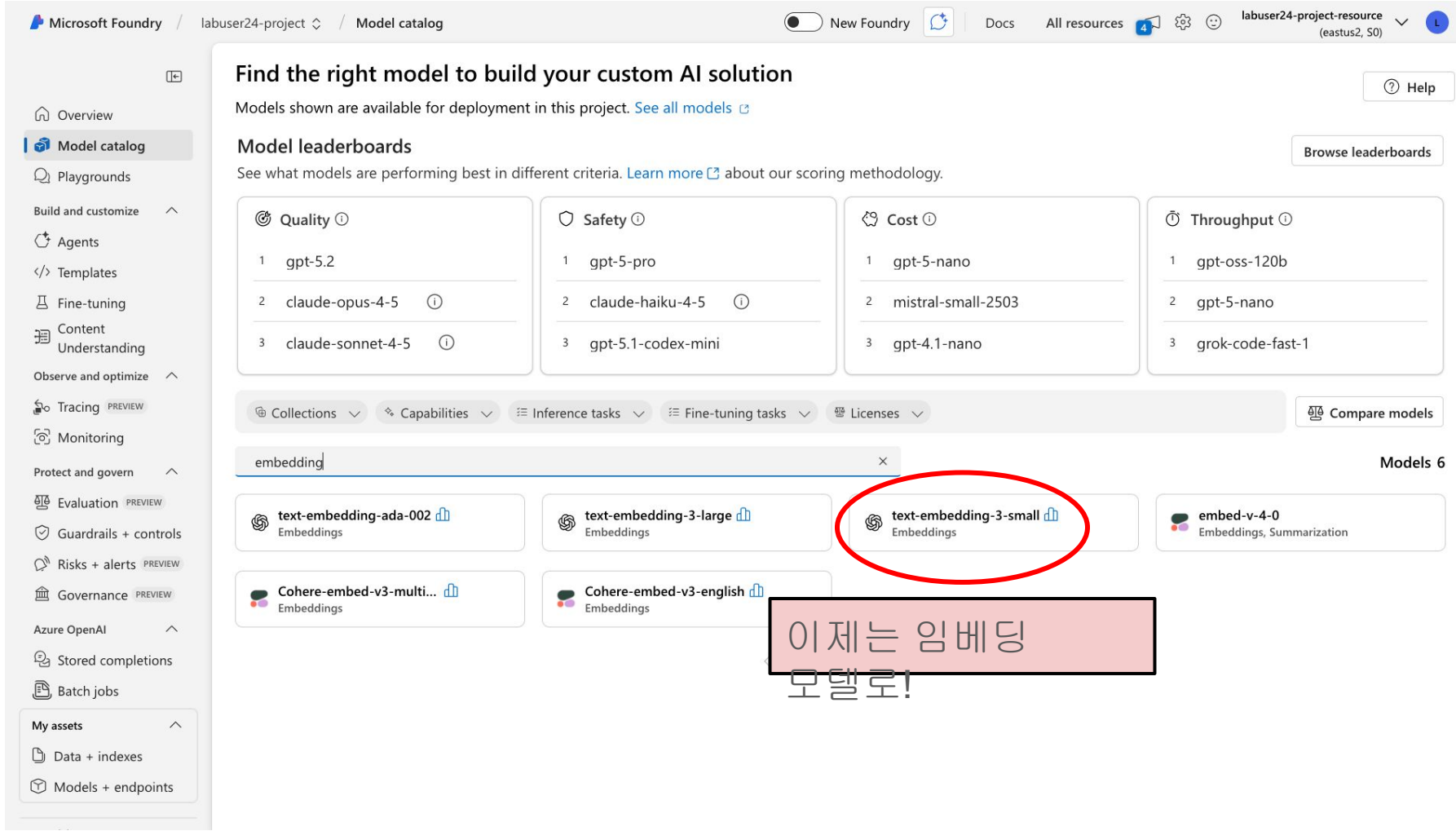
02.loader.py 파일에서 여러가지 로더들 실험!
(/Users/jaebeomyim/PythonProjects/Cloud_Computing/02.loader.py)

tiktoken이라는 모듈은 토큰의 수를 세어주는 패키지.
왜 필요하냐면 pdf 파일을 RAG 할 때, 파일의 용량이 너무 크면 나눠줘야 하는데 (splitter), 이때 토큰 수를 잘 분해하기 위해서...?

splitter로는 거의 Recursive만 씀!! (비교는 뒷 페이지 참조.)

- OpenAI API만 쓰면 내 로컬에 있는 PDF 파일이나 엑셀 데이터를 모델에게 먹이기가 매우 번거롭습니다 (텍스트를 다 긁어서 프롬프트에 붙여넣어야 함).
- **LangChain**은 `Document Loader` 라는 도구가 있어서, PDF, Word, 웹사이트, 유튜브 자막 등을 모델이 이해하기 쉬운 형태로 쪼개고(Chunking), 저장하고(Vector Store), 검색해서(Retriever) 가져오는 **전 과정을 자동화**해 줍니다.

종류	작동 방식	장점	단점
RecursiveCharacter (추천)	문단 > 문장 > 단어 순으로 **의미 단위**를 지키며 자름	문맥이 잘 유지됨. RAG 성능이 가장 좋음.	연산이 아주 미세하게 더 걸림 (무시 가능 수준).
Character (기본)	그냥 지정한 구분자(예: `\n\n`) 하나로만 자름	단순하고 빠름.	구분자가 없으면 무식하게 잘라서 **단어가 짤릴 수 있음.**
TokenTextSplitter	LLM이 읽는 **토큰(Token) 개수** 기준으로 자름	LLM의 입력 한도(Context Window)를 정확히 맞출 때 유리.	문장의 의미보다는 토큰 수에 집착해서 문맥이 끊길 수 있음.
CodeSplitter (Python, JS 등)	프로그래밍 언어의 문법(클래스, 함수 등)을 기준으로 자름	**코드 분석**에 최적화.	일반 텍스트(소설, 뉴스)에는 부적합.

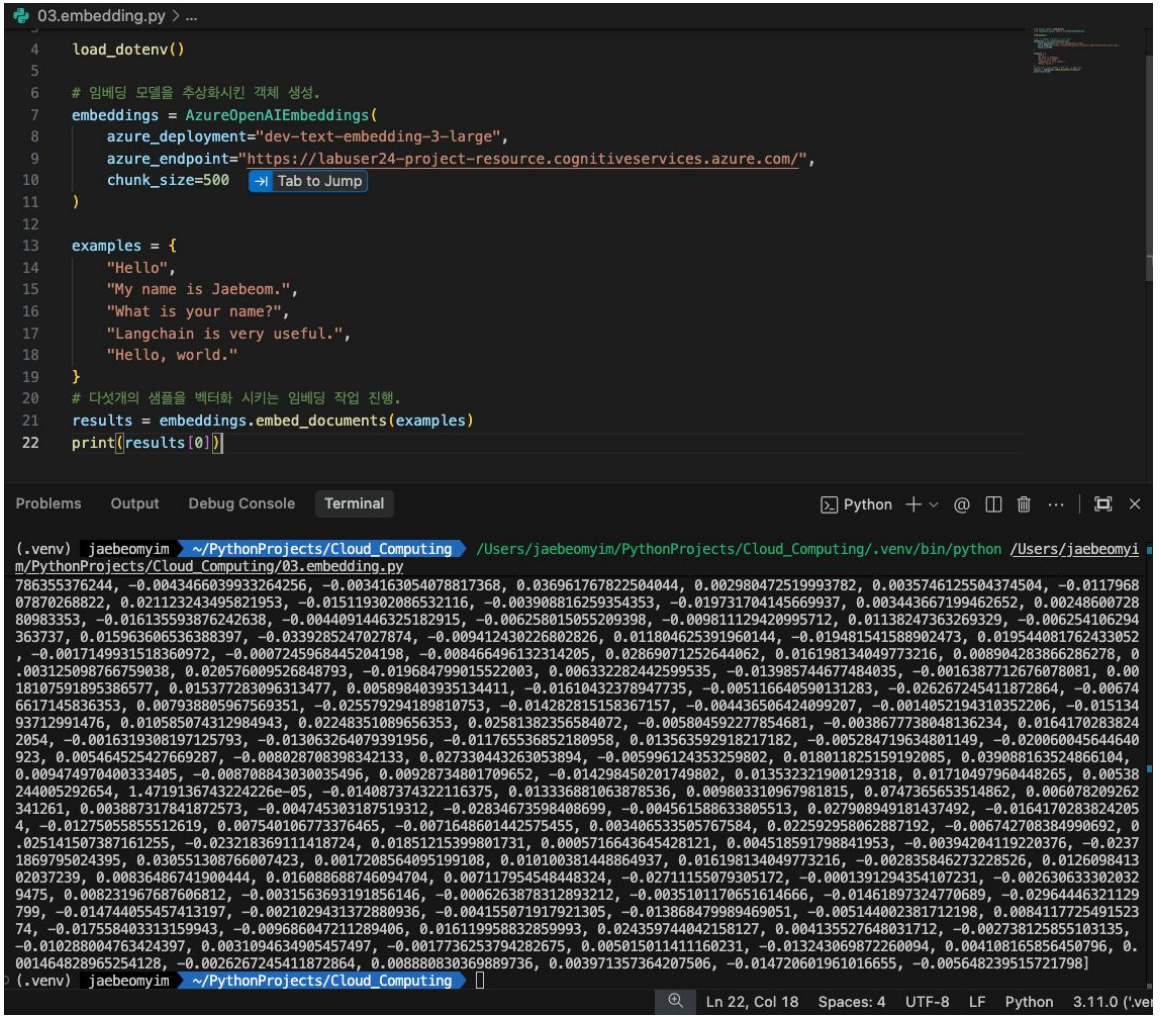

Microsoft Foundry
labuser24-project
Model catalog
New Foundry
Docs
All resources
labuser24-project-resource
(eastus2, S0)
Overview
Model catalog
Playgrounds
Build and customize
Agents
Templates
Fine-tuning
Content Understanding
Observe and optimize
Tracing PREVIEW
Monitoring
Protect and govern
Evaluation PREVIEW
Guardrails + controls
Risks + alerts PREVIEW
Governance PREVIEW
Azure OpenAI
Stored completions
Batch jobs
My assets
Data + indexes
Models + endpoints
Find the right model to build your custom AI solution
Models shown are available for deployment in this project. See all models
Help
Model leaderboards
See what models are performing best in different criteria. Learn more about our scoring methodology.
Browse leaderboards
Quality
1 gpt-5.2
2 claude-opus-4-5
3 claude-sonnet-4-5
Safety
1 gpt-5-pro
2 claude-haiku-4-5
3 gpt-5.1-codex-mini
Cost
1 gpt-5-nano
2 mistral-small-2503
3 gpt-4.1-nano
Throughput
1 gpt-oss-120b
2 gpt-5-nano
3 grok-code-fast-1
Collections
Capabilities
Inference tasks
Fine-tuning tasks
Licenses
Compare models
embedding
Models 6
text-embedding-ada-002
Embeddings
text-embedding-3-large
Embeddings
text-embedding-3-small
Embeddings
embed-v-4-0
Embeddings, Summarization
Cohere-embed-v3-multi...
Embeddings
Cohere-embed-v3-english
Embeddings
이제는 임베딩 모델로!

Embedding도 모델이 따로있구나...
이런 임베딩 모델, llm 들이 모여 체인을 이룰 것으로 예상

벡터 방식 임베딩 모델을 쓰면 문자열/문서/쿼리 등을 N차원 공간의 벡터로 바꿔줌 (N=15xx, in our example). 어떻게 변환하는지는 모델에 따라 다름.

그래서 문자열들 간의 cosine similarity를 계산할 수 있게 됨!

(03.embedding.py 파일 참고.)



Overall Process of constructing a vector db and retrieving.

1. Load data. (PyPDFLoader(...))
1. Split data into chunks. (RecursiveCharacterTextSplitter(...))
2. Embed the data into vector format. (AzureOpenAIEmbeddings(...))
3. Save in a vector database. (Chroma.from_documents(...))
4. Retrieve the most relevant segment via similarity search. (db.similarity_search(...))

```
(.venv) jaebeomyim ~/PythonProjects/Cloud_Computing /Users/jaebeomyim/PythonProjects/Cloud_Computing/.venv/bin/python /Users/jaebeomyim/PythonProjects/Cloud_Computing/04.vectordb.py
2.2 Hilbert spaces
Hilbert spaces are defined as follows.
Definition 2.2. An inner product vector space His called a Hilbert space if it is complete (relative to the
norm associated to the scalar product).
We know that the Banach spaces given in the previous sub-section can be equipped with a scalar product
when p= 2 and this provides us with examples of Hilbert spaces.
2.2.1 Examples of Hilbert spaces
• The inner product vector spaces Rn and Cn with inner product (1.56) are complete (since they are
Banach spaces relative to the norm associated to this scalar product) and they are, hence, Hilbert
spaces. We know that any finite-dimensional inner product vector space over the field R (over the
field C) is isomorphic to Rn (or Cn) by mapping a vector to its coordinate vector relative to some
chosen basis. If we choose an ortho-normal basis we know from Eq. (1.32) that, in terms of the
coordinates, the scalar product can be expressed in terms of the standard scalar product on Rn or
Cn. Together, these facts imply that any finite-dimensional inner product vector space over R or C
is a Hilbert space.
• For the measure set ( X,Σ,μ), the space L2(X), defined in Eq. (1.90) is an inner product vector
space with inner product given by Eq.(1.92). We already know that this is a Banach space (relative
to the norm associated to the scalar product), so L2(X) is complete and, hence, a Hilbert space.
• Associated to the measure space ( N,Σc,μc) with counting measure μc we have the space ℓ2 of all
sequences (xi)∞
i=1 in R (or C) with
(∑∞
i=1 |xi|2)1/2 finite. An inner product on this space is given by
Eq. (1.97). Since ℓ2 is a Banach space it is complete and is, hence, also a Hilbert space.
```

수리물리학 렉쳐노트를 벡터db화 해서 올려놓은 다음 “What is Hilbert space?”라고 물어본 결과.

